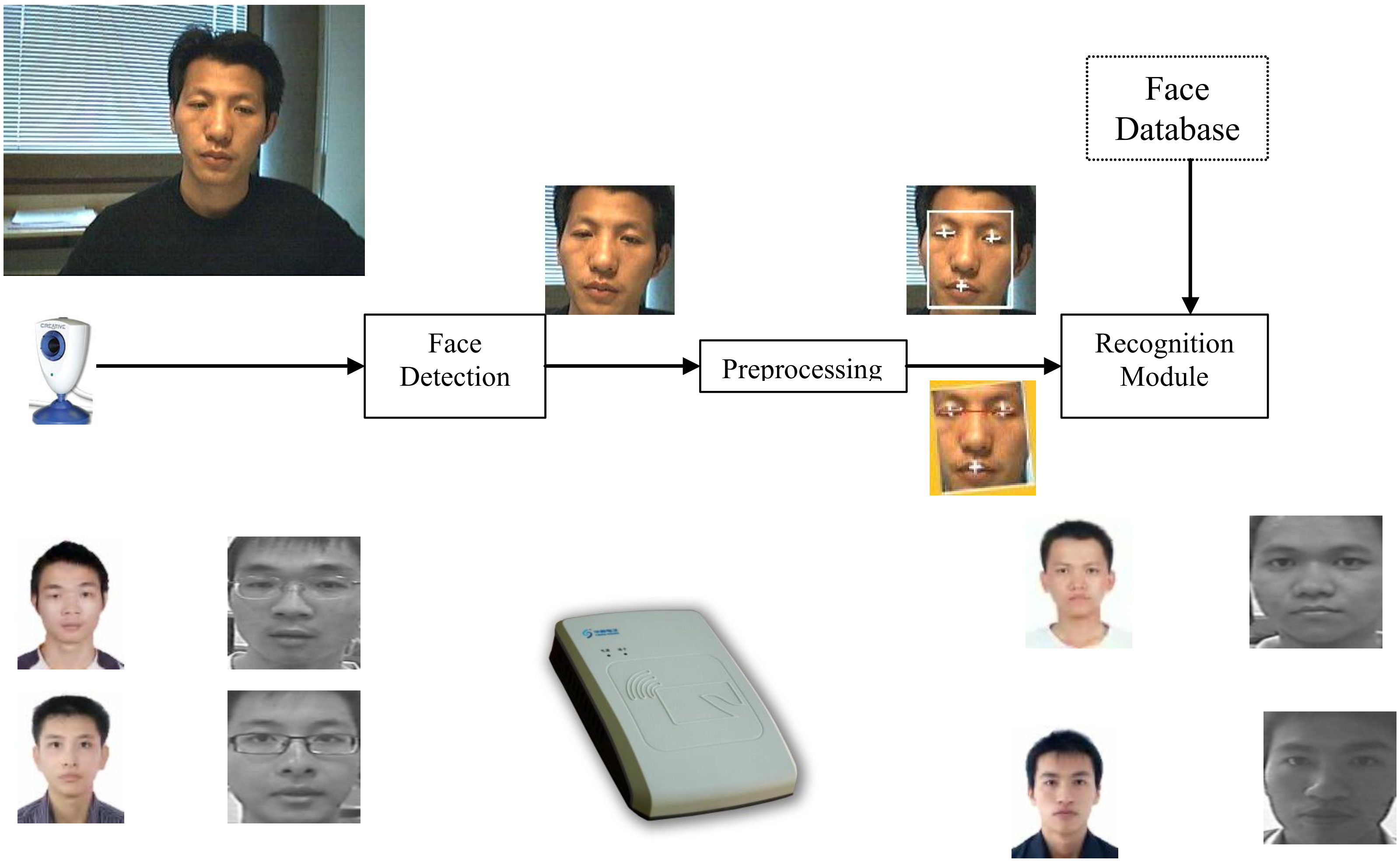
人脸识别

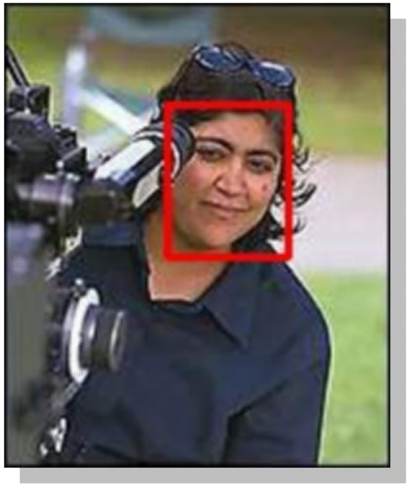
System

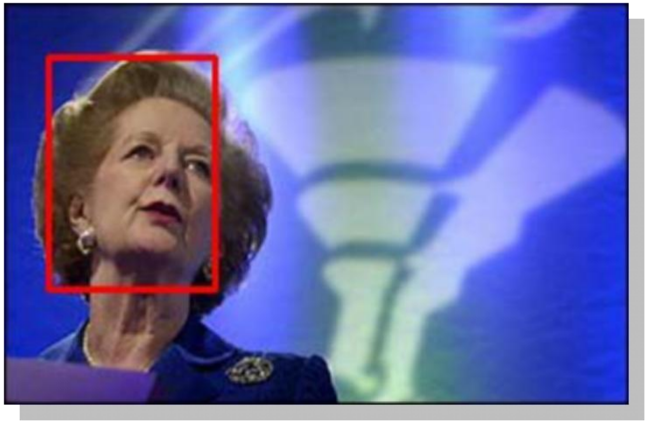


Face Database
Face Detection
Preprocessing
Recognition Module

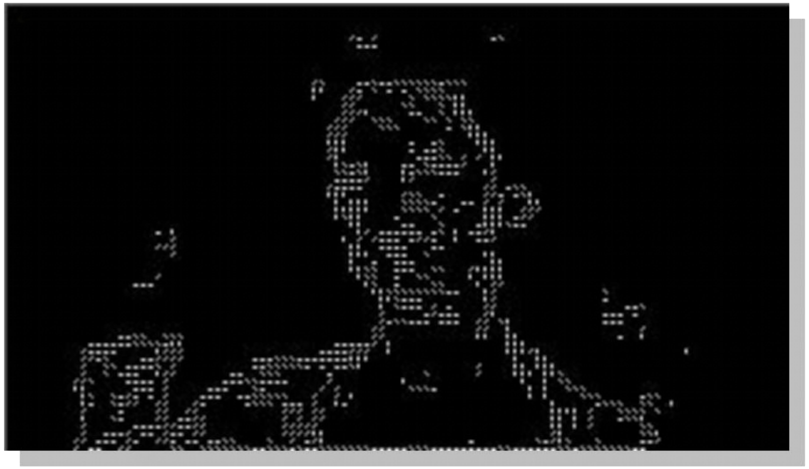
Segmentation- Face Detection

- Detect/locate the faces given an image
  - •Edge, color, template based
  - •Facial components based
  - •Classification based

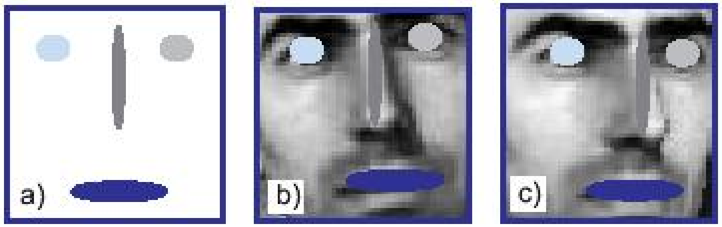


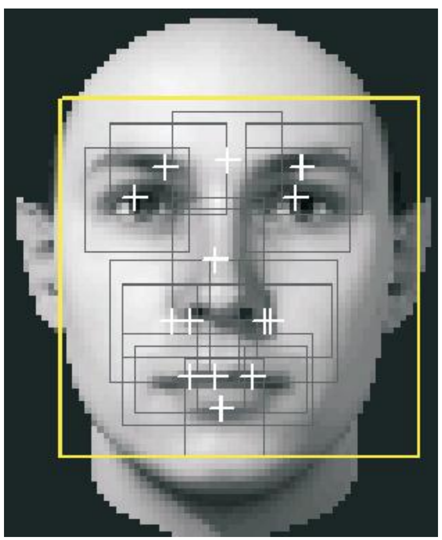


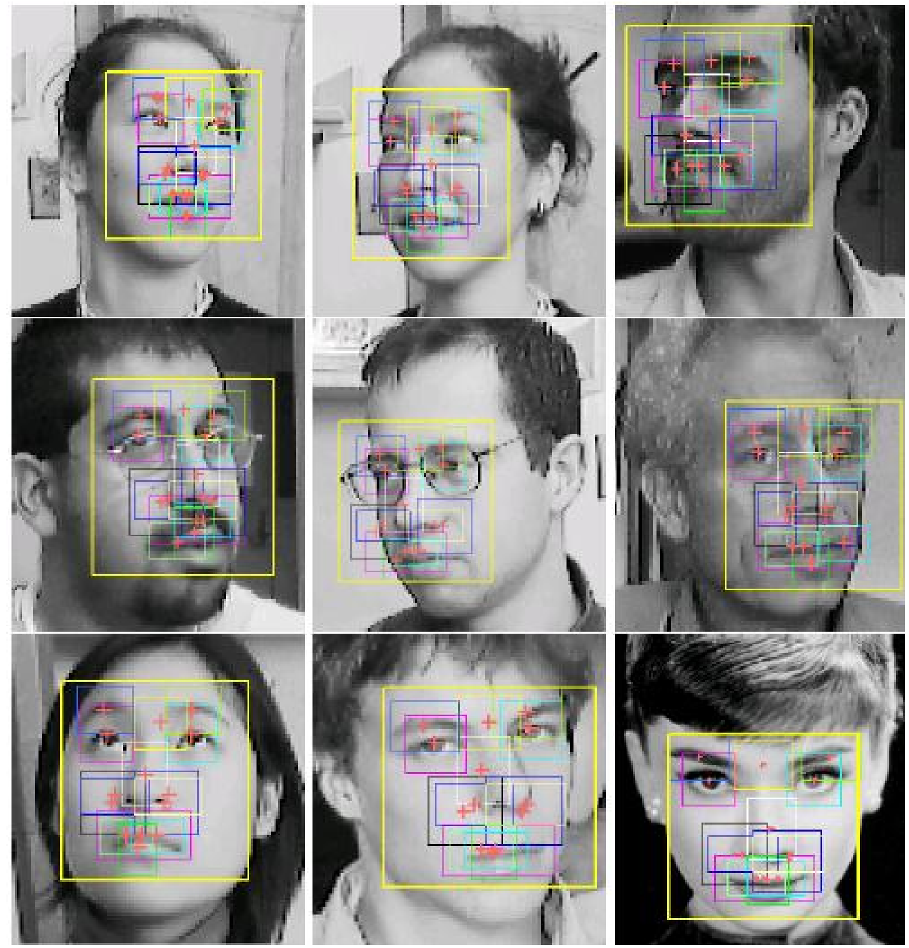
Edge and/or Color Based



Components Based

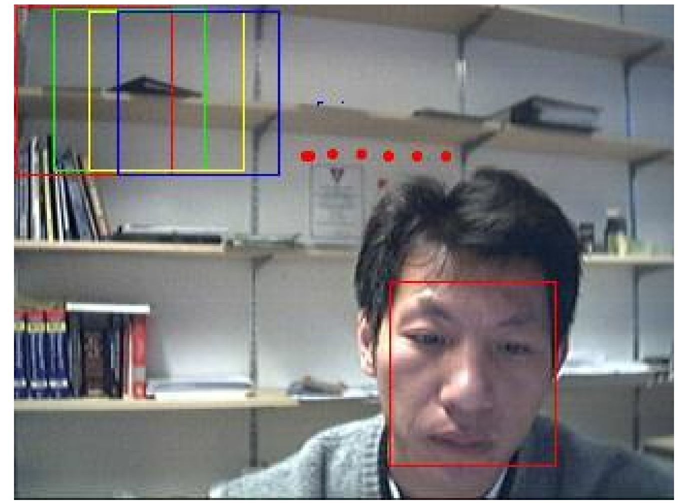




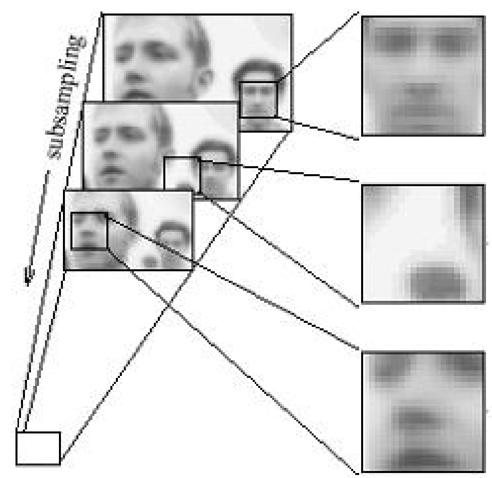


Classification Based

- Train a face/non-face classifier
- Scan an image with a window
- Input the window to the classifier



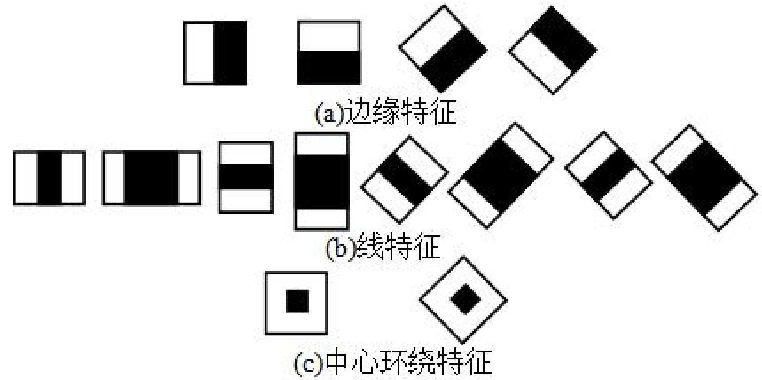
Input image pyramid



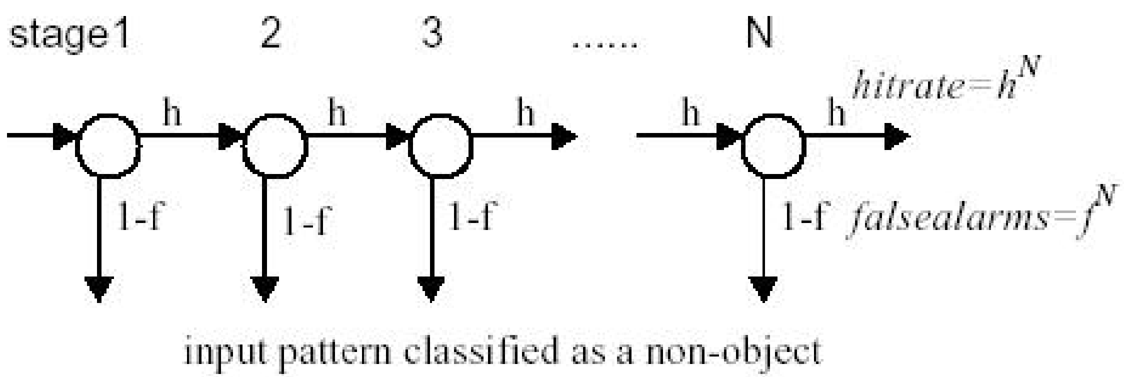
What about the size of the window?

75,081,800 windows to be scanned!

State of the Art Face Detection Algorithm



- Viola & Jones (2001)
- Haar-like feature
  - Very fast computation
- Boosting based face/non-face classifier
  - Cascade of classifiers, very efficient
- 100ms for an image with size: 320*240



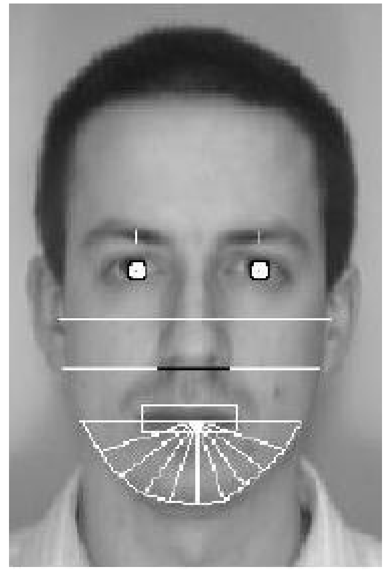
Preprocessing

- illumination normalization
  - Histogram Equalization
- Geometric normalization
  - Rotation
- Scale normalization
  - Size

Feature Extraction

- Geometric feature based
- Appearance based
- Local feature based
- Hybrid

Geometric Features



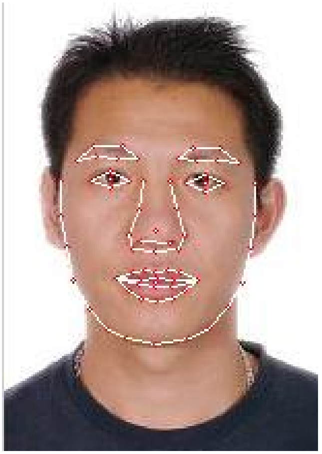
- Brunelli & Poggio, 1993
- Nose width, length, chin shapes
- Bayes classifier

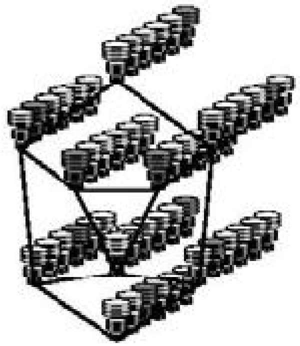
Appearance Based

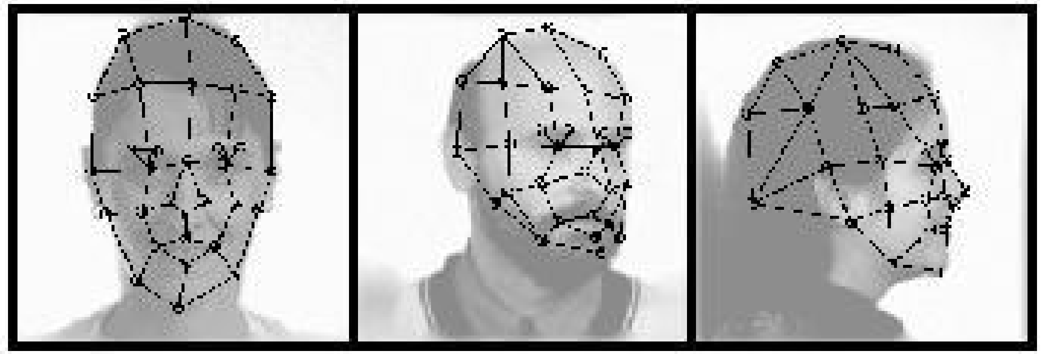
Measure of the differences
Or Correlations

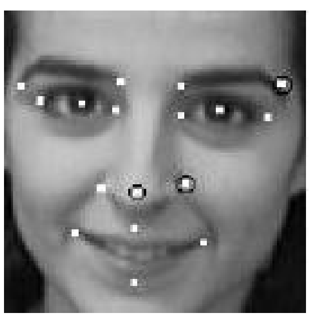
- Eigenface, Fisherface
  - Distance between features

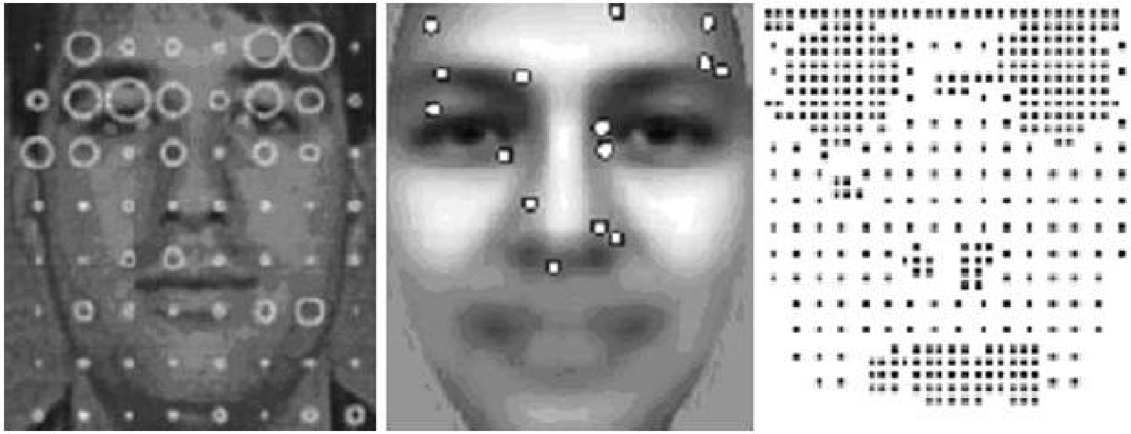
Local Feature Based

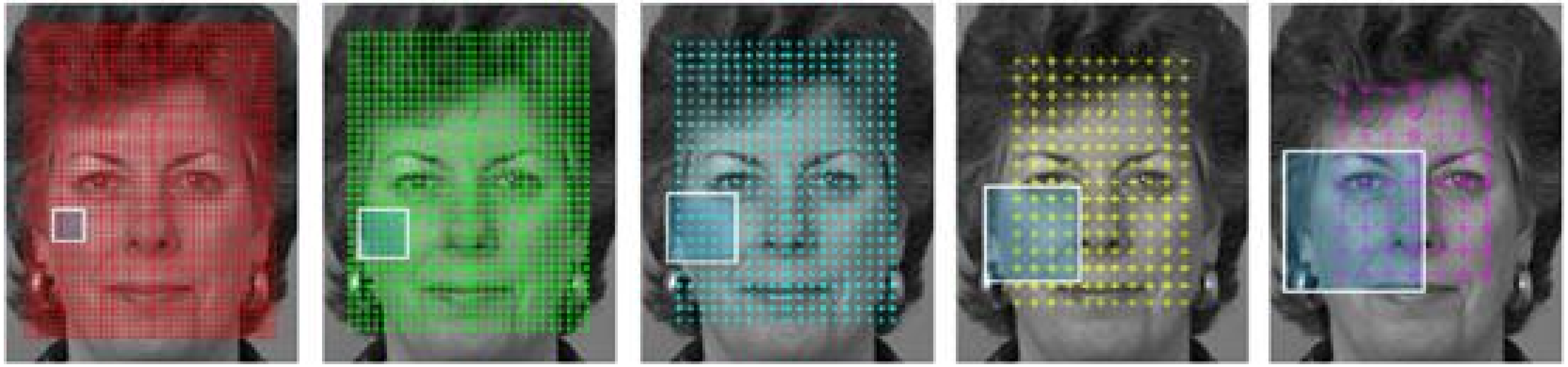












Hybrid

- Fusing local and global feature
- Feature fusion
- Classifier fusion

FERET Evaluation

Probe Set	Gallery	Probe set size	Gallery size	Variations
Fb	Fa	1195	1196	Expression
Fc	Fa	194	1196	Illumination and Camera
Dup I	Fa	722	1196	Time gap < 1 week
Dup II	Fa	234	1196	Time gap > 1 year

fa

fb

duplicate I

fc

duplicate II

Test Results (96/97)

Methods	**Fb**	**Fc**	**Dup I**	**Dup II**
Baseline_PCA	79.0%	18.9%	40.8%	22.0%
Baseline_Correlation	83.0%	10.0%	36.6%	18.0%
Rutgers (Greyscale Projection)	76.1%	18.8%	31.2%	20.9%
MIT (PCA)	95.0%	32.0%	58.1%	32.5%
MSU (Fisher)	88.3%	30.0%	32.5%	10.4%
UMD (Fisher)	**96.2%**	59.0%	48.1%	21.3%
USC (DLA)	95.0%	**82.0%**	**59.1%**	**52.1%**

Progress in FERET Test

Methods	**Fb**	**Fc**	**Dup I**	**Dup II**
DLA (1997)	95.0%	82.0%	59.1%	52.1%
LBP (2005)	93.0%	51.0%	61.0%	50.0%
LGBHS (2005)	94.0%	97.0%	68.0%	53.0%
Selected Gabor + Kernel (2006)	96.7%	85.6%	59.3%	62.4%
HGPP (2007)	97.6%	**98.9%**	77.7%	76.1%
Fusing Gabor and LBP (2007)	**98.0%**	98.0%	**90.0%**	**85.0%**

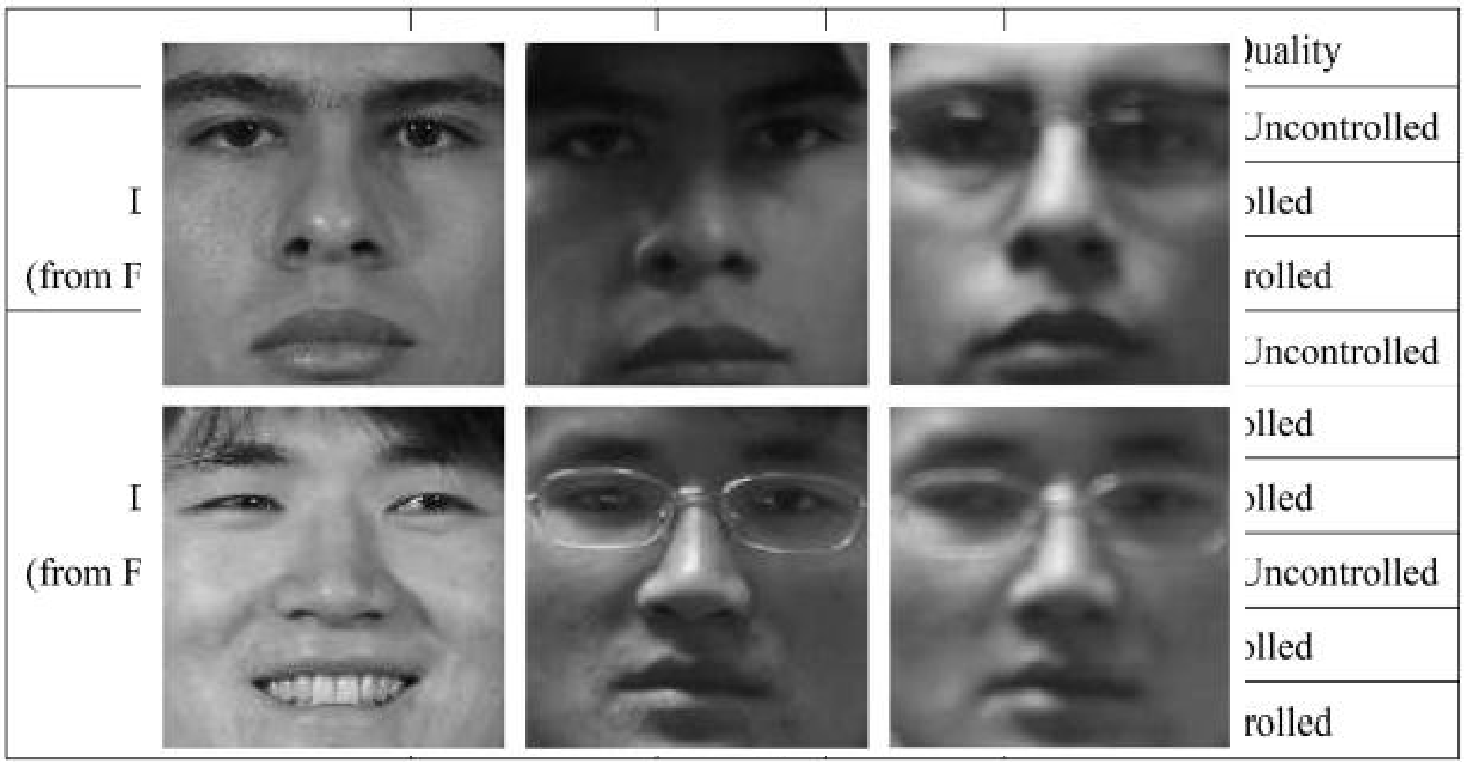
Progress in FERET Test

Methods	**Fb**	**Fc**	**Dup I**	**Dup II**
LGPDP (2008)	97.3	97.2	79.8	77.5
LGBP_Mag + LGXP (2010)	99.0	99.0	94.0	93.0
LMG (2011)	99.8	99.5	89.2	86.8
GV-LBP-TOP-P (2011)	97.99	98.97	81.86	83.76
E-GV-LBP-M (2011)	98.41	98.97	81.99	81.62
Gabor Surface (2011)	**99.6**	**99.5**	**94.0**	**91.5**

BANCA Test 2004

- The BANCA database
  - 52 subjects, 12 sessions, 10 shots per session
  - Different camera, pose, backgrounds
- Testing protocols
  - Specified training and test images
  - Weighted Error Rate (WER)
- The participants
  - About 12 from the world, e.g. Surry, CMU, Tsinghua

Face Recognition Grand Challenge (FRGC)



Performance

Methods	FRGC, FAR=0.1%		
	ROC I	**ROC II**	**ROC III**
Kernel LDA with fractional power polynomial kernel (06)	-	-	76
Fusing Gabor and LBP (07)	-	-	83.6
Improved LGPDP (2009)	82.82	80.74	78.36
LGBP_Mag + LGXP (2010)	83.9	84.7	85.2
E-GV-LBP-M + P (2011)	**88.71**	**89.38**	**89.99**

Progress in Face Recognition

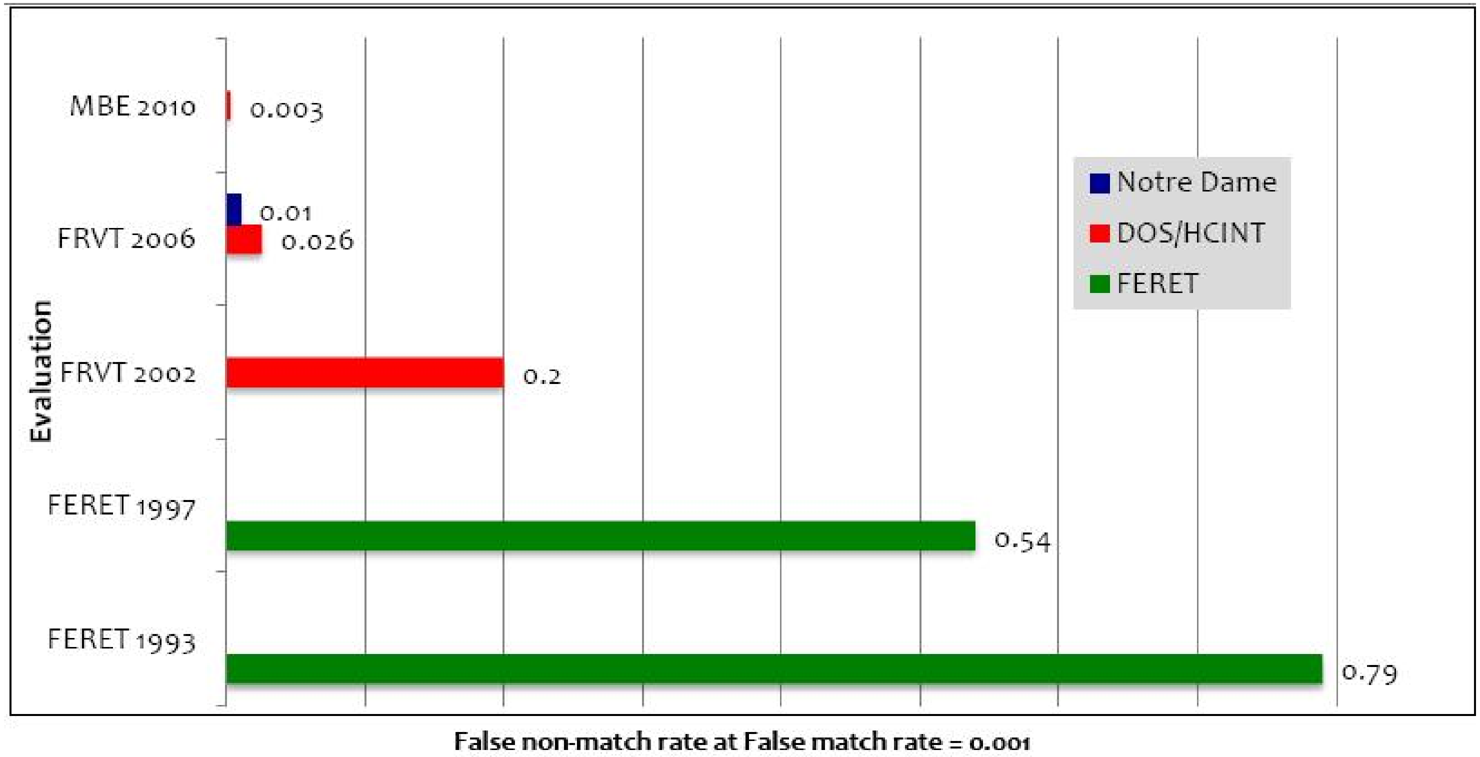
NEC, L1 Identity

Neven Vision

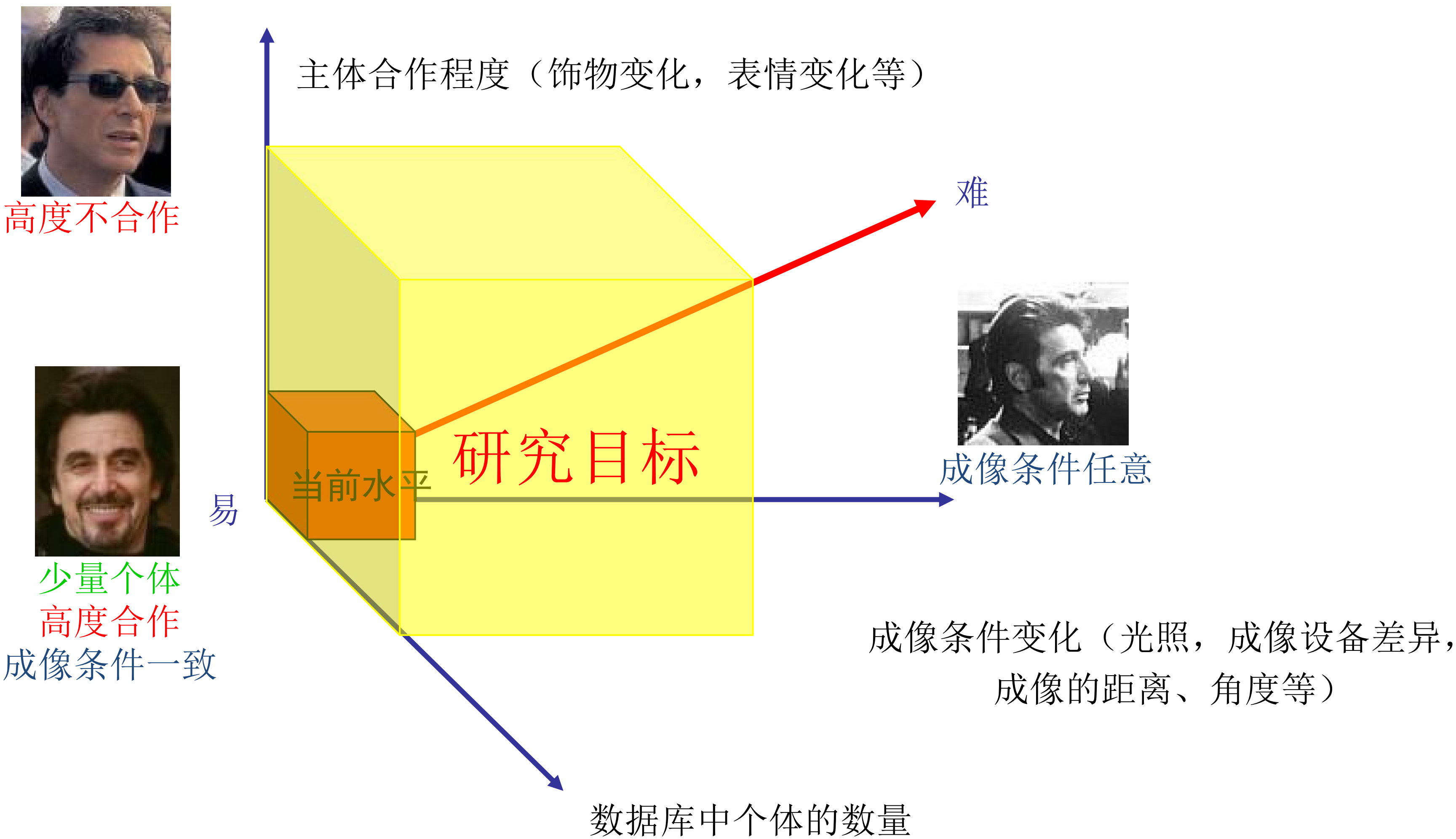
Cognitec, Eyematic

Elastic Graph

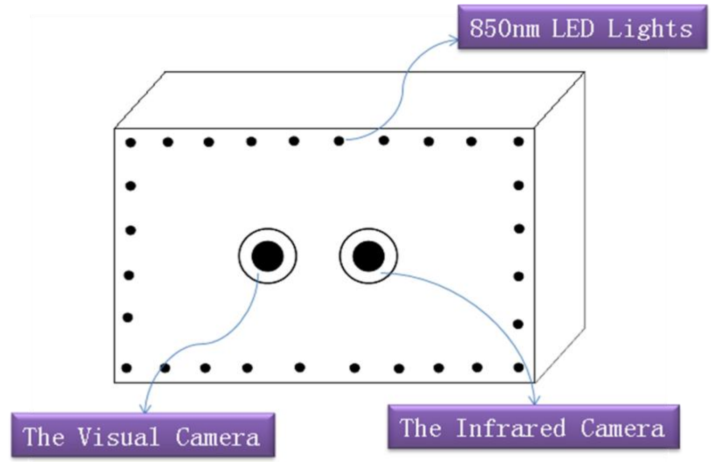
Eigenface



Challenges

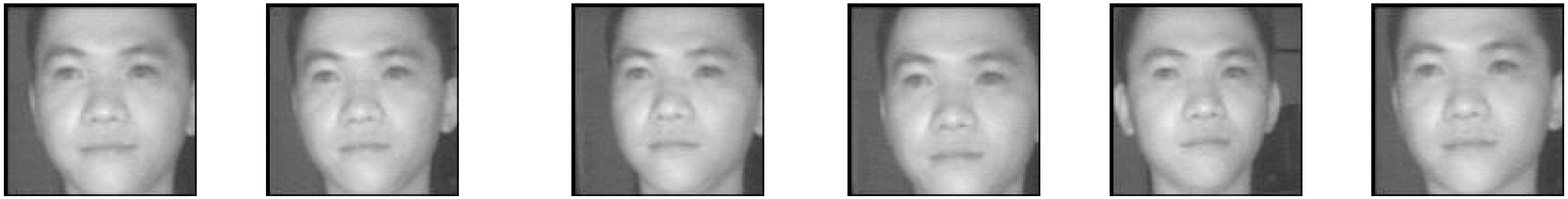

主体合作程度（饰物变化，表情变化等）
高度不合作
难
研究目标
当前水平
易
成像条件任意
少量个体
高度合作
成像条件一致
成像条件变化（光照，成像设备差异，
成像的距离、角度等）
数据库中个体的数量

近红外人脸

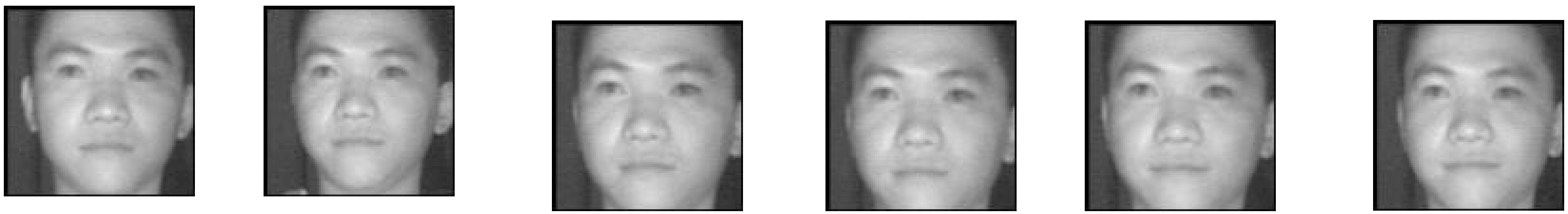


Peak wavelength λp=850nm

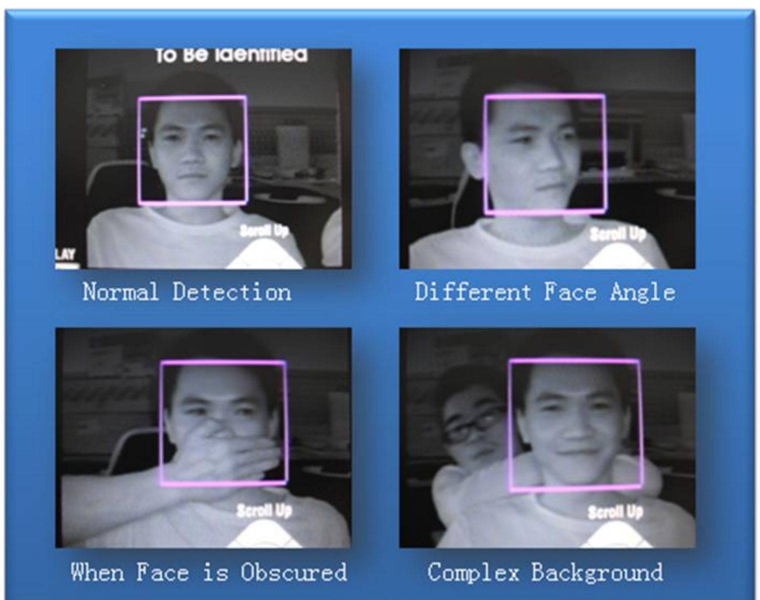
- Image Sensor: 1/3-inch CCD Image Sensor
- CCD Total Pixels: 795(H)*596(V)
- TV System: PAL

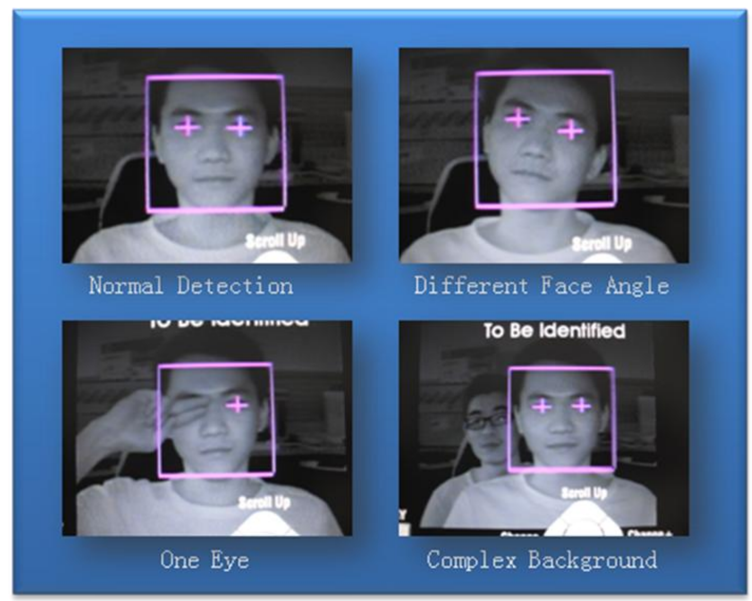


Example face images captured when lights are on

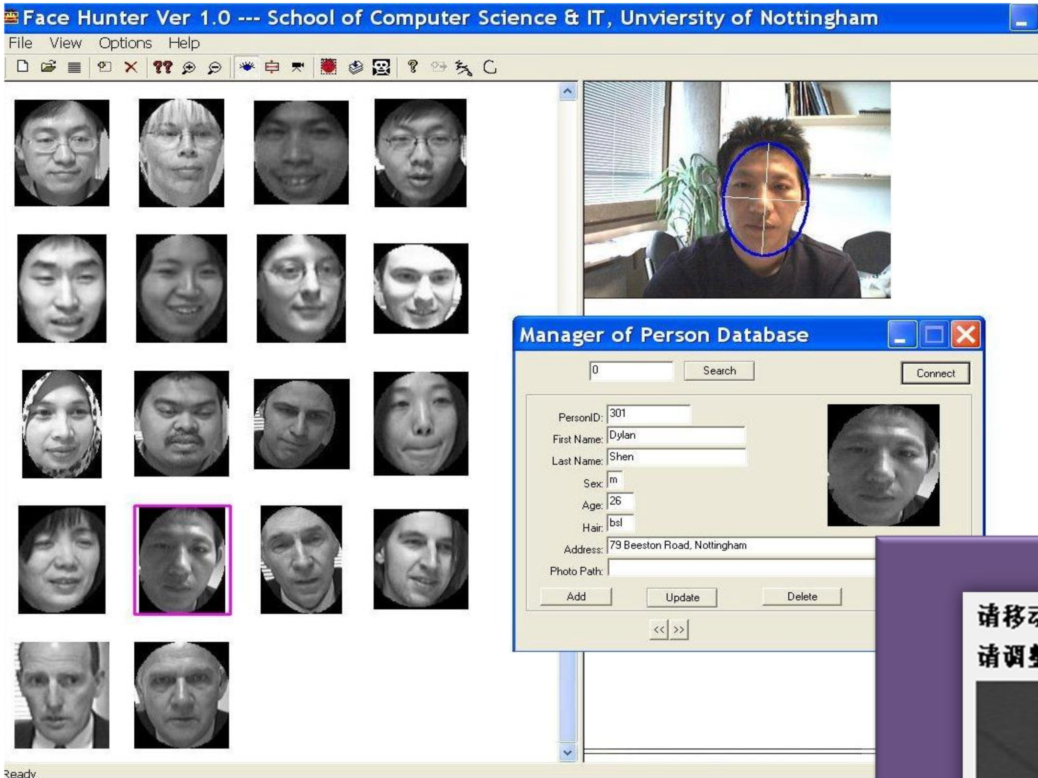


Example face images captured when lights are off





- 300人
  - 红外图像 +可见光
  - 身份证照片－１１５人
  - 两个session（间隔时间〉１月）
    - ７０人



PC上近红外人脸识别示例


TI DSP CONTEST
2008 TI DSP 大奖赛
DSP CONTEST

三 等 奖
（算法实现组）

参赛学校：深圳大学
项目名称：基于Gabor小波的人脸识别算法及其在达芬奇平台上的实现
指导教师：沈琳琳
团队队员：蔡晶 张烨妮 周

Texas Instruments Technology fo


2008 年德州仪器
大陆-台湾两岸 DSP 大奖赛邀请赛
算法实现组

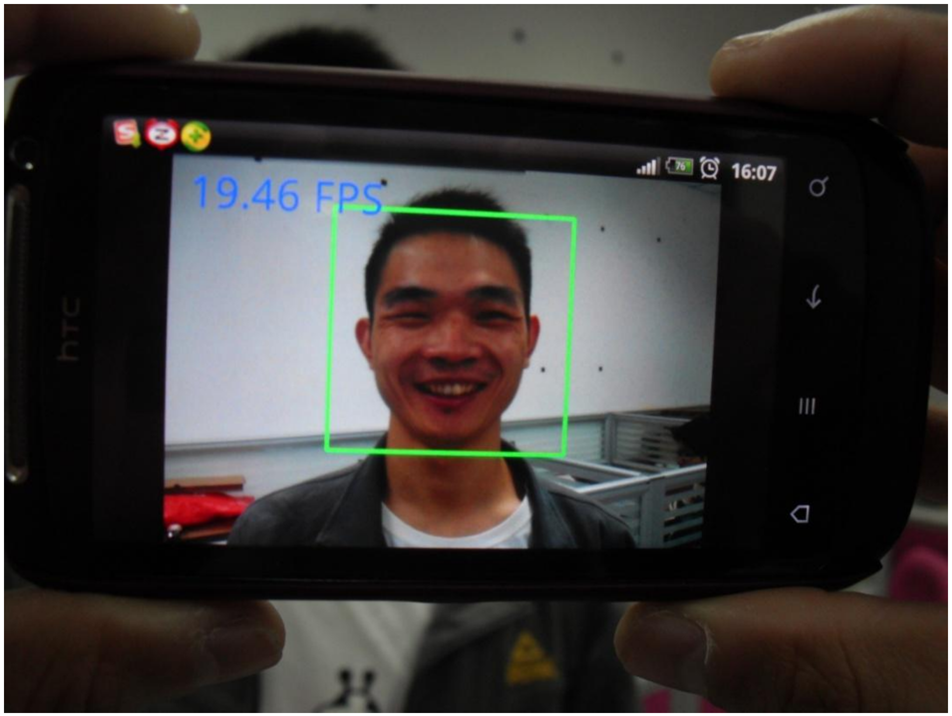
TI DSP 2008 Design Challenge Final

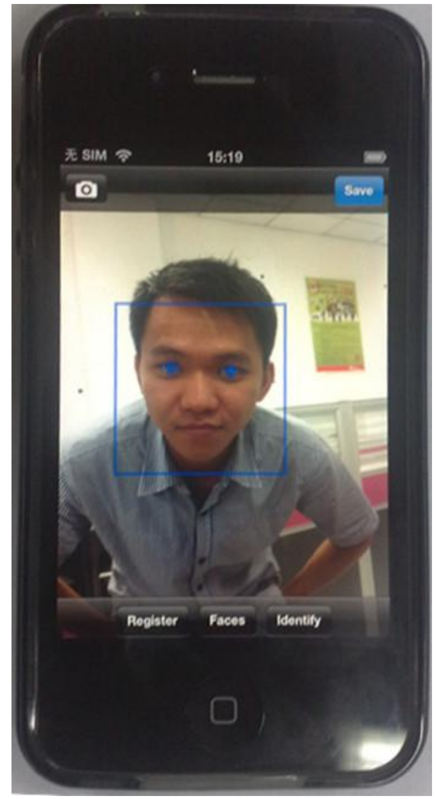
三 等 奖

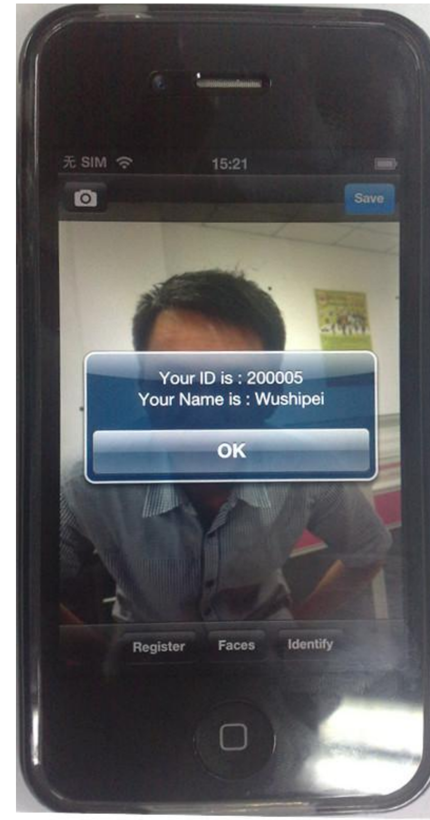
参赛队伍：深圳大学
项目名称：基于 Gabor 小波的人脸识别算法及其在达芬奇平台上的实现
指导老师：沈琳琳
团队成员：周家锐　张烨妮

Texas Instruments
德州仪器半导体技术（上海）有限公司
二零零八年十一月

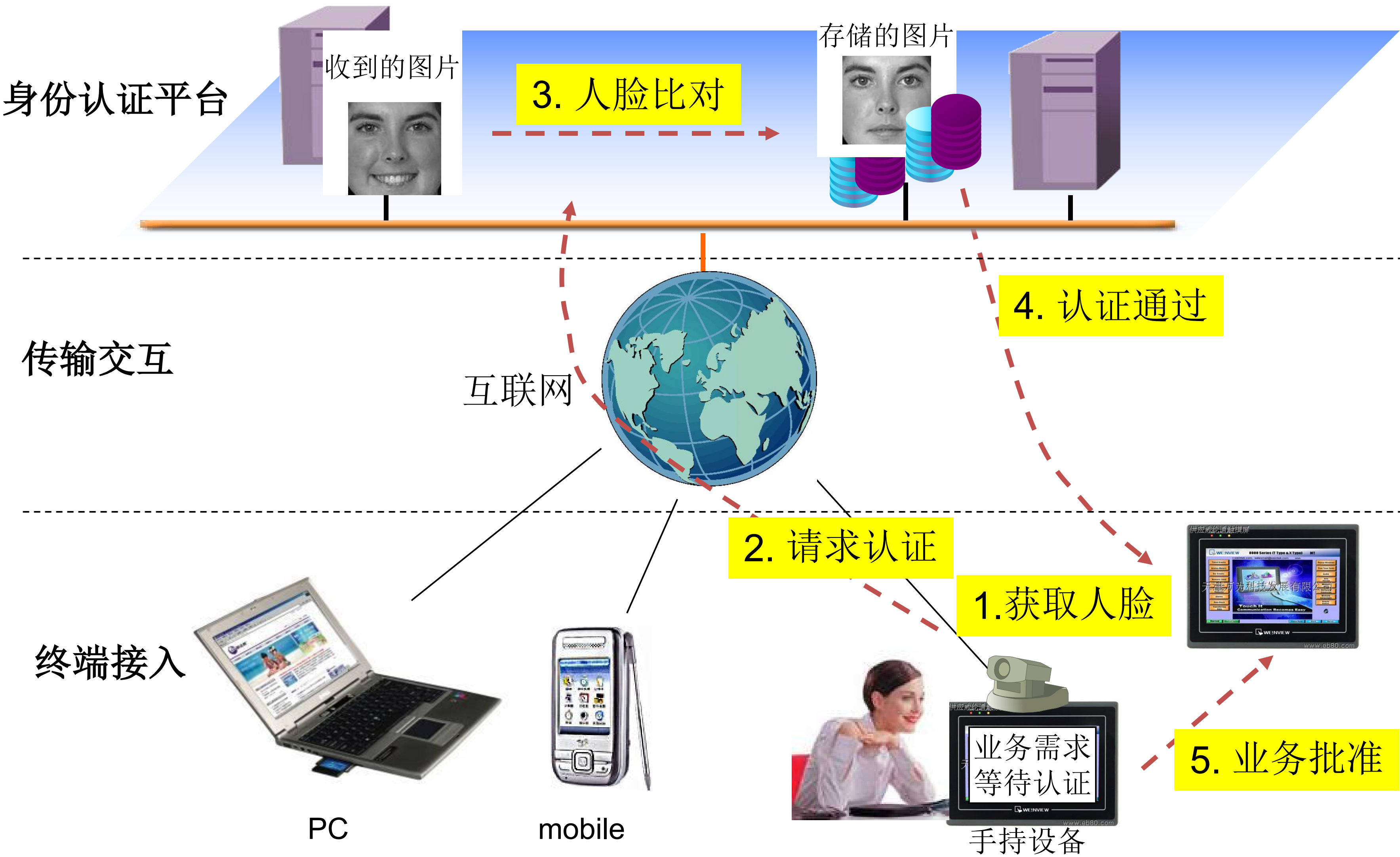
Mobile Phone Based System







互联网身份认证



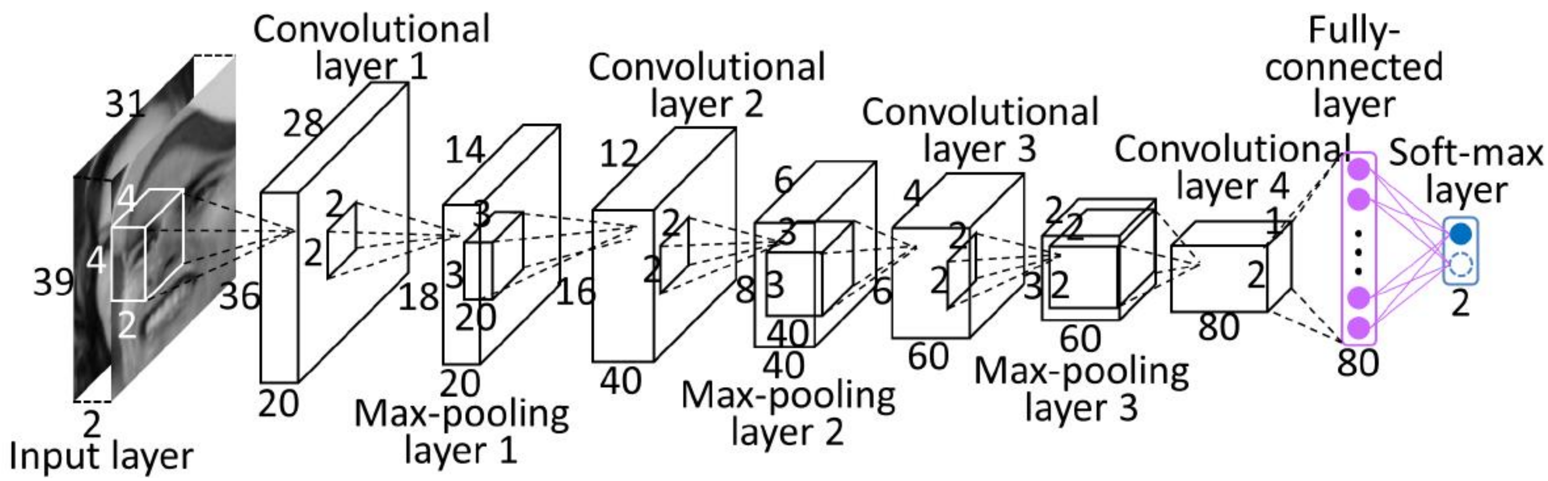
身份认证平台
收到的图片
3. 人脸比对
存储的图片
传输交互
互联网
4. 认证通过
终端接入
2. 请求认证
1.获取人脸
业务需求
等待认证
5. 业务批准
PC
mobile
手持设备

Learn Identity Features with Verification Signal

- Extract relational features with learned filter pairs

$$y^j = f\left(b^j + k^{1j} * x^1 + k^{2j} * x^2\right)$$

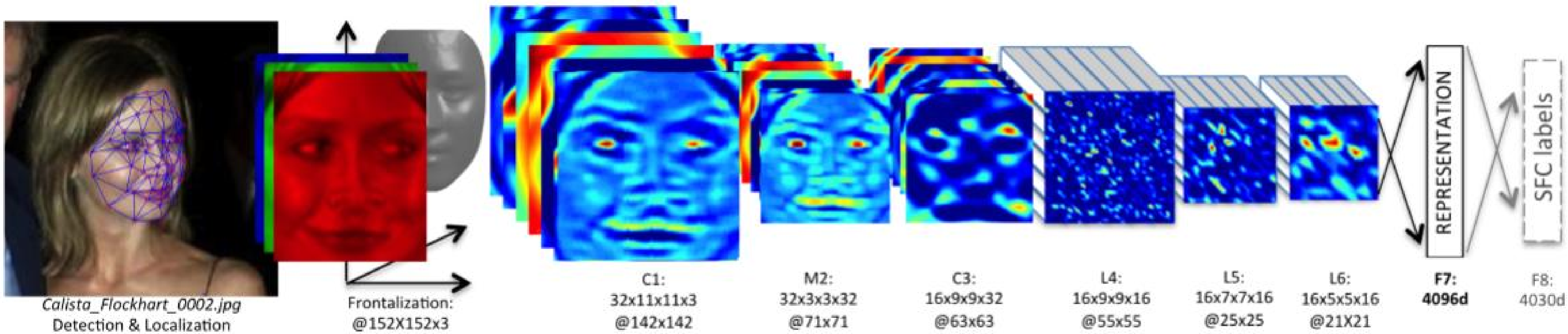
- These relational features are further processed through multiple layers to extract global features
- The fully connected layer can be used as face representations



Y. Sun, X. Wang, and X. Tang, “Hybrid Deep Learning for Computing Face Similarities,” Proc. ICC

• DeepFace(Facebook)

- **Outline of the DeepFace architecture**

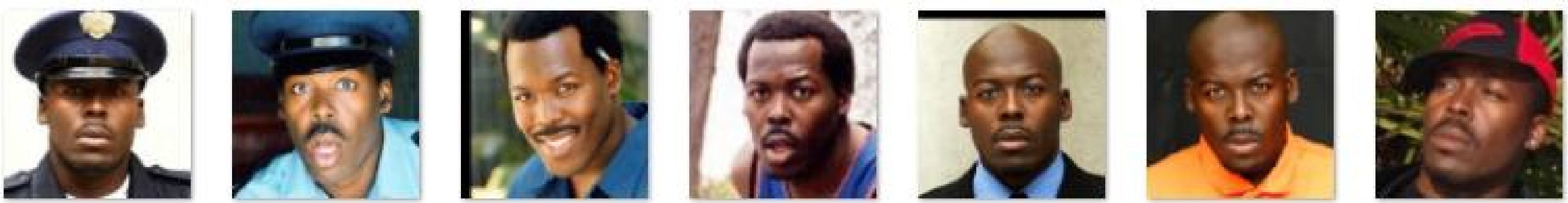


- A front-end of a single convolution-pooling-convolution filtering on the rectified input, followed by three locally-connected layers and two fully-connected layers. Colors illustrate feature maps produced at each layer. The net includes more than 120 million parameters, where more than 95% come from the local and fully connected layers.

- Taigman, Y., Yang, M., Ranzato, M., Wolf, L.: Deepface: Closing the gap to humanlevel performance in face verification. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 1701–1708 (2014)

CASIA-WebFace

- Collected from Internet using a semi-automatical way.
- Containing 10,575 subjects and 494,414 images.
- Average: 47 images per subject.
- Link: http://www.cbsr.ia.ac.cn/english/CASIA-WebFace-Database.html .



Yi, Dong, et al. "Learning face representation from scratch." arXiv preprint arXiv:1411.7923 (2014).

MS-Celeb-1M

- Obtained from Freebase based on their occurrence frequencies (popularities) on the web.
- Containing 99,952 celebrities and 10,490,534 images.
- Average: 105 images per entity.
- Link: https://www.microsoft.com/en-us/research/project/ms-celeb-1m-challenge-recognizing-one-million-celebrities-real-world/ .

Guo, Yandong, et al. "Ms-celeb-1m: A dataset and benchmark for large-scale face recognition." European Conference on Computer Vision. Springer International Publishing, 2016.

Progress in FERET Test

Methods	**Fb**	**Fc**	**Dup I**	**Dup II**
LGPDP (2008)	97.3	97.2	79.8	77.5
LGBP_Mag + LGXP (2010)	99.0	99.0	94.0	93.0
LMG (2011)	99.8	99.5	89.2	86.8
GV-LBP-TOP-P (2011)	97.99	98.97	81.86	83.76
E-GV-LBP-M (2011)	98.41	98.97	81.99	81.62
Gabor Surface (2011)	99.6	99.5	94.0	91.5
Statistical local feature (2013)	**99.7**	**99.5**	**96.3**	**94.4**
VGG	99.7	98.5	94.6	95.3
SZU	**99.83**	**100**	**99.58**	**99.57**

Labeled Faces In the Wild Dataset (LFW)

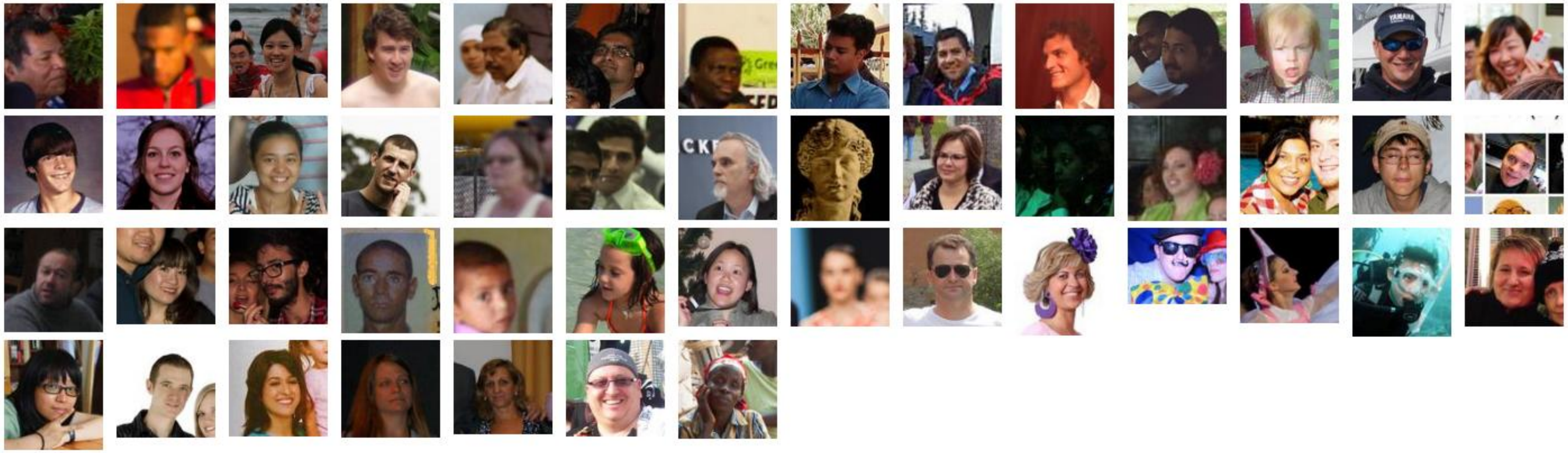
- Face Verification: Given a pair of images specify whether they belong to the same person
- 13K images, 5.7K people
- Standard benchmark in the community
- Several test protocols depending upon availability of training data within and outside the dataset.

Comparison with the State of the Art

No.	Method	# Training Images	# Networks	Accuracy
1	Fisher Vector Faces	-	-	93.10
2	DeepFace	4 M	3	97.35
3	DeepFace Fusion	500 M	5	98.37
4	DeepID-2,3	Full	200	99.53
5	FaceNet	200 M	1	98.87
6	FaceNet+ Alignment	200 M	1	99.63
7	VGG	2.6 M	1	98.95
8	NTechLab			98.10
9	Face++			99.50
10	Tencent BestImage			99.65
11	Baidu			99.77
12	DaHua			99.78

MegaFace

- Obtained from Flickr (Yahoo's dataset).
- Containing 690,572 identities and 1,027,060 images (Challenge 1).
- Containing 672,057 identities and 4.7 Million images (Challenge 2).
- Link: http://megaface.cs.washington.edu/



Kemelmacher-Shlizerman, Ira, et al. "The megaface benchmark: 1 million faces for recognition at scale." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016.

MegaFace

Challenge 1

More details:

http://megaface.cs.washington.edu/results/

Algorithm	Date Submitted	Set 1	Set 2	Set 3
BingMMLab V1(iBUG cleaned data)	4/10/2018	98.998%	98.998%	98.998%
Orion Star Technology (clean)	3/21/2018	[illegible]		
iBUG_DeepInsight	2/8/2018	98.063%	[illegible]	98.053%
EM-DATA	4/4/2018	96.653%	96.653%	96.653%
SuningUS_AILab	3/21/2018	[illegible]	[illegible]	96.2618947%
StartDT-AI	4/16/2018	93.8226%	93.8226%	93.8226%
Intellivision	2/11/2018	93.125%	93.123%	93.136%
NTechLAB - facenx_large	10/20/2015	73.3%	73.309%	73.287%
ForceInfo	04/07/2017	72.11%	72.084%	72.121%
3DiVi Company - tdvm V2	04/15/2017	71.742%	71.727%	71.703%
DeepSense - Small	07/31/2016	70.983%	70.948%	70.962%
Google - FaceNet v8	10/23/2015	70.496%	70.492%	70.551%

人类：23.9%

Kemelmacher-Shlizerman, Ira, et al. "The megaface benchmark: 1 million faces for recognition at scale." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016.

网红人脸识别

同一个人？ 否

网红人脸识别

同一个人？　　否

1

2

3

4

5

6

7

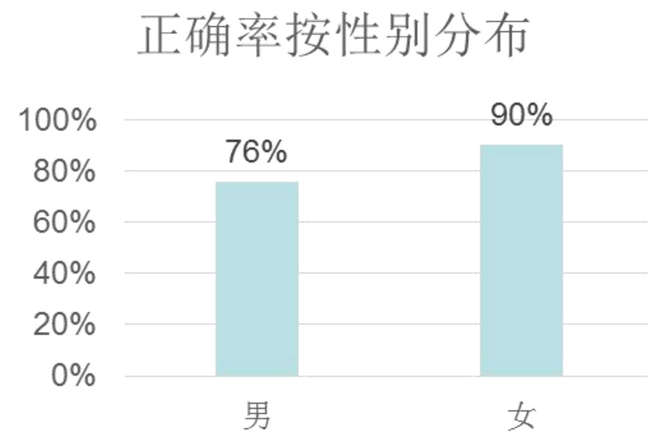
8

9

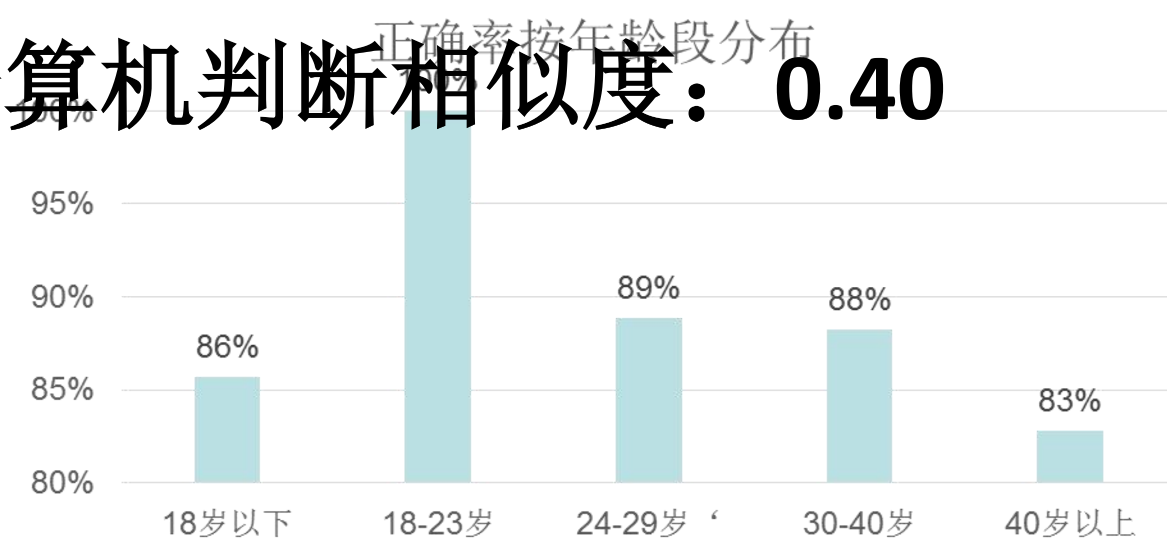
10

241人测试

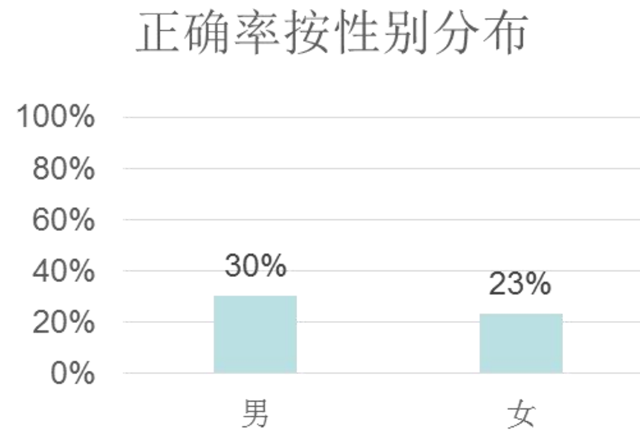
正确率最高（87%），不同人

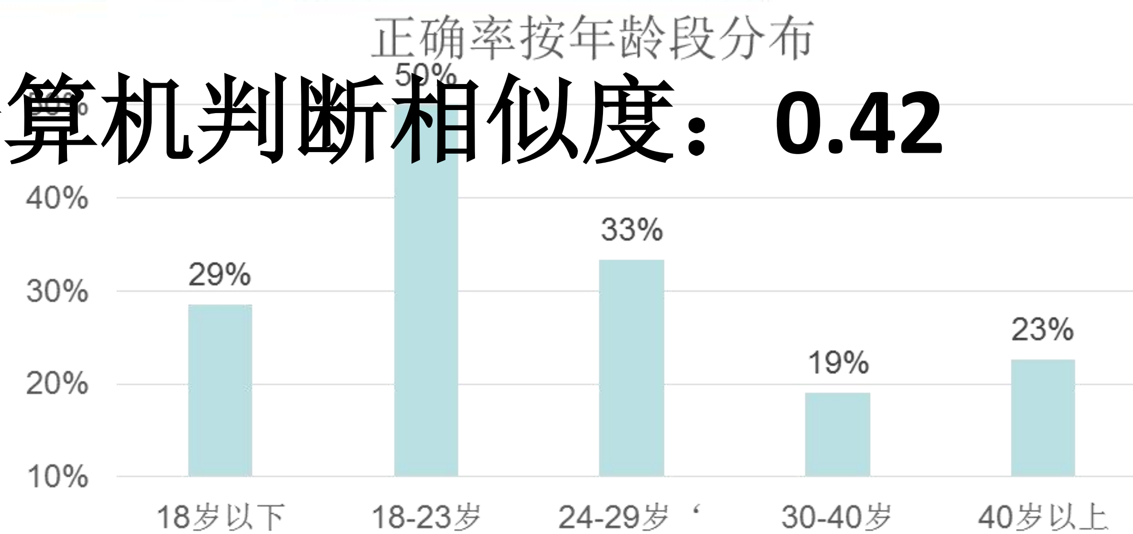


计算机判断相似度：0.40



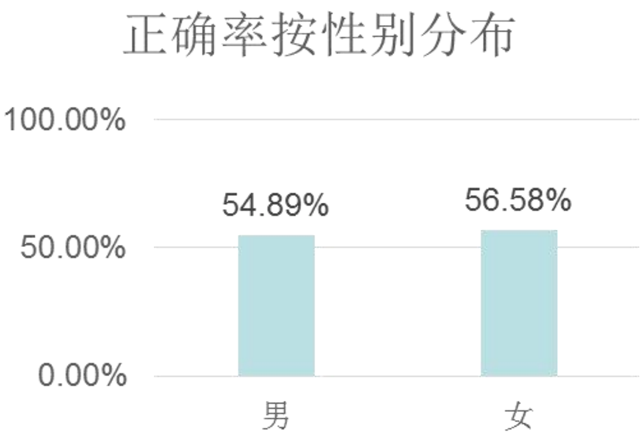
正确率最低（26%）-不同人

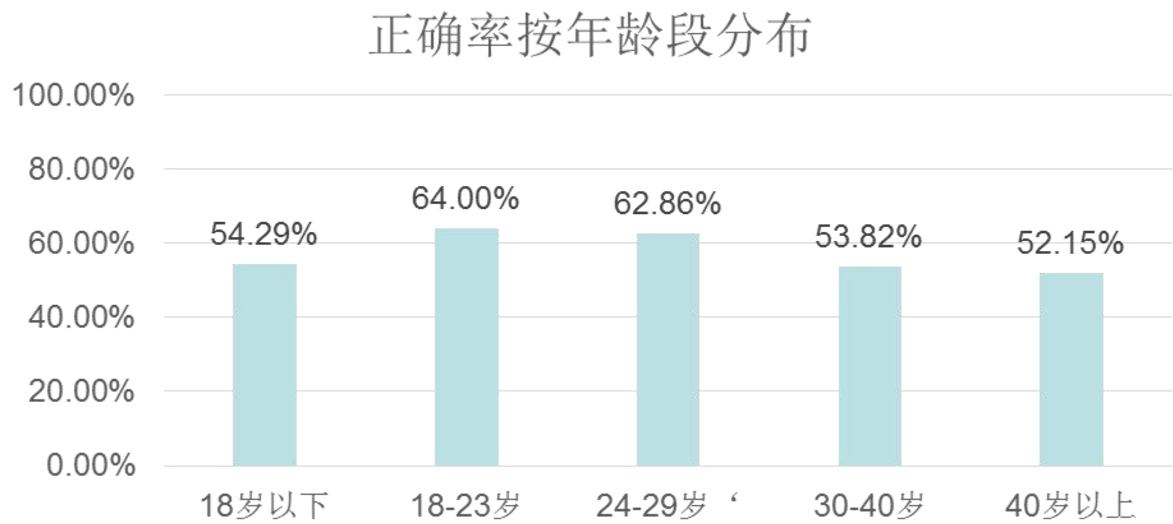




计算机判断相似度：0.42

人和电脑的比较





人	55.7%
深度网络A	**80%**
深度网络B	60%
非深度学习算法	51%

Same or Different face?

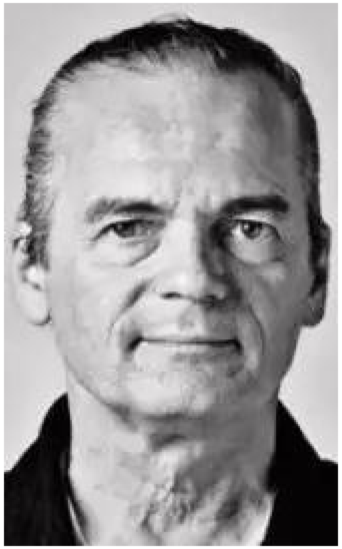
Angelababy

Angelababy

First 4： DCNN correct， Students wrong
The 5th： Students correct, DCNN wrong

The first 4 image pairs are from Similar-Looking LFW database

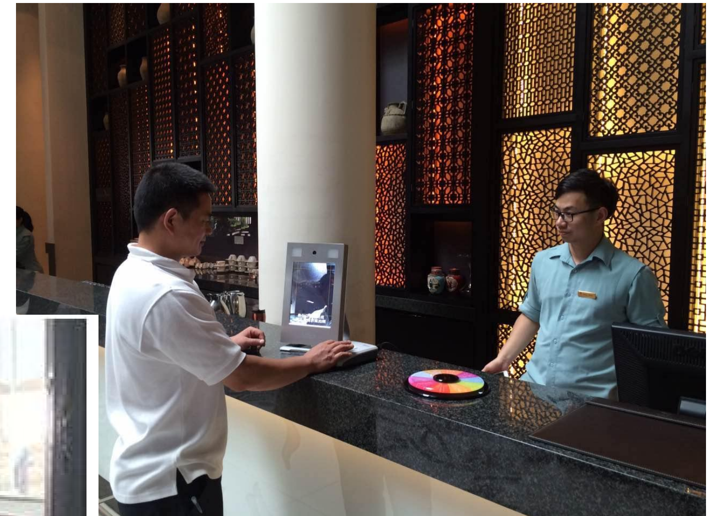
FRANÇOIS BRUNELLE



地球上有人和你长一样！

人证核查

- 杭州G20
- 深圳文博会、高交会、双创周
- 新疆



人证核查

- Test on Standard Id Pairs


请您刷卡
风干肉
味道自然就好!

刷脸取款（2017.9）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刷 脸 取 款
Pay 取 款
无卡预约取款
WITHDRAWAL WITHOUT CARD
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CONTACTLESS SERVICE
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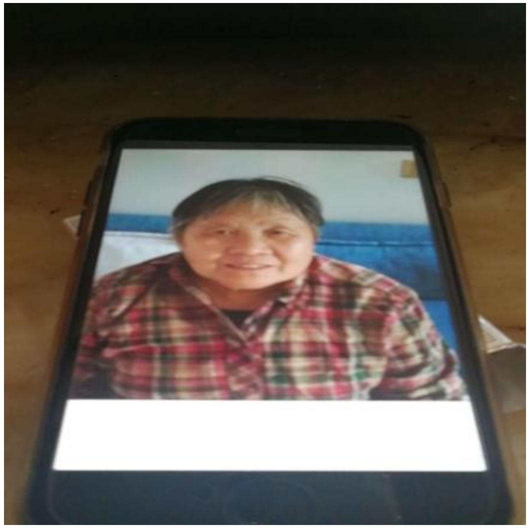
50cm
3. 请与摄像头保持50CM内距离
确认/Ok
取消/Cancel
www.abchina.com

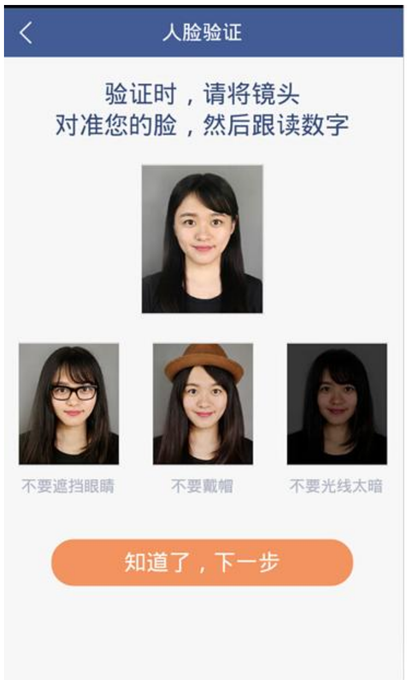

ABC 中国农业银行
95599
54
请输入手机号或身份证号码
1
2
3
4
5
6
7
8
9
00
0
.
取消
Cancel
更正
Correct
确认
Confirm
更 正
确 认
取 消
刷身份证
提示：字母用“.”键代替。
www.abchina.com

Spoofing 欺骗攻击

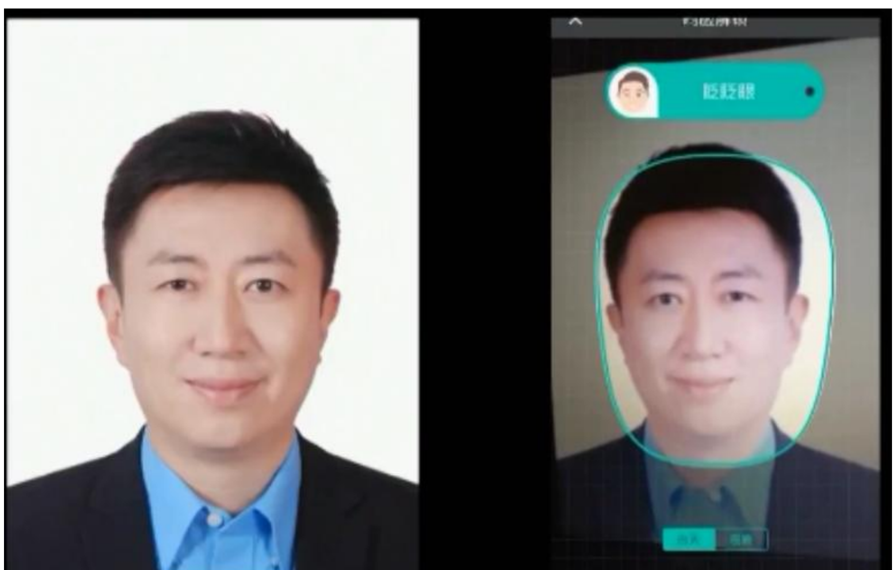
- 相片攻击： 打印的照片
- 视频攻击： 播放视频



Anti-Spoofing 反欺骗



Voice

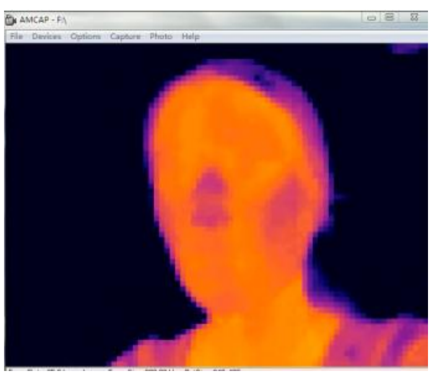


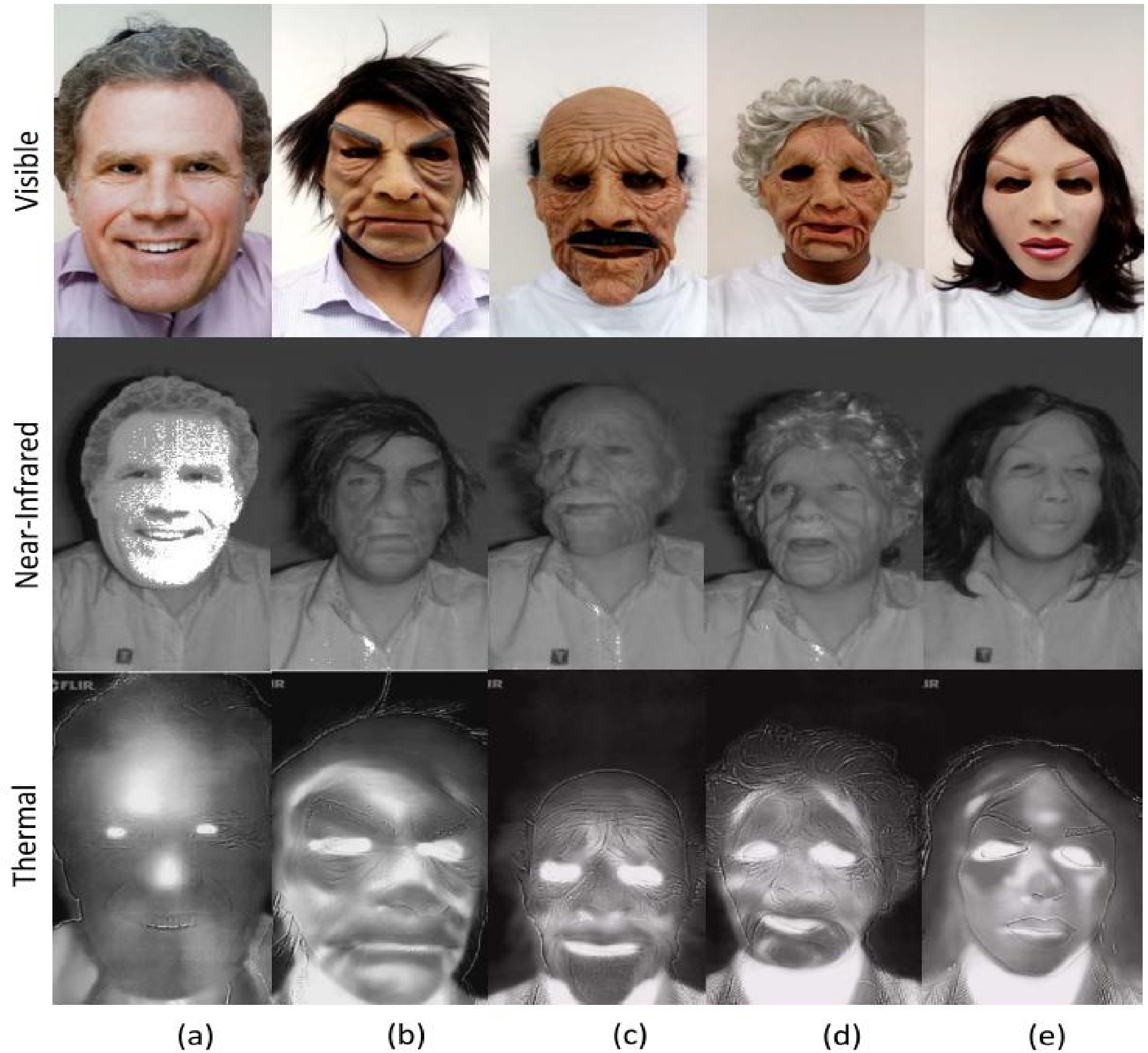
照片欺骗（眨眼）

视频欺骗（摇头）


DOT PROJECTOR
7MP CAMERA
MICROPHONE
AMBIENT LIGHT SENSOR
FLOOD ILLUMINATOR
PROXIMITY SENSOR
INFRARED CAMERA

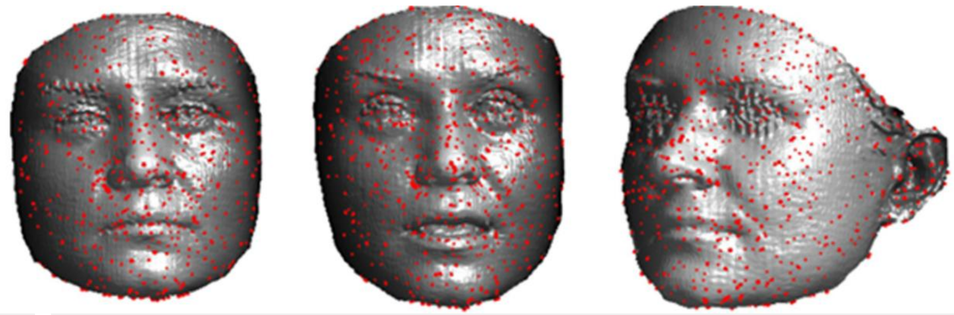
3D 面具

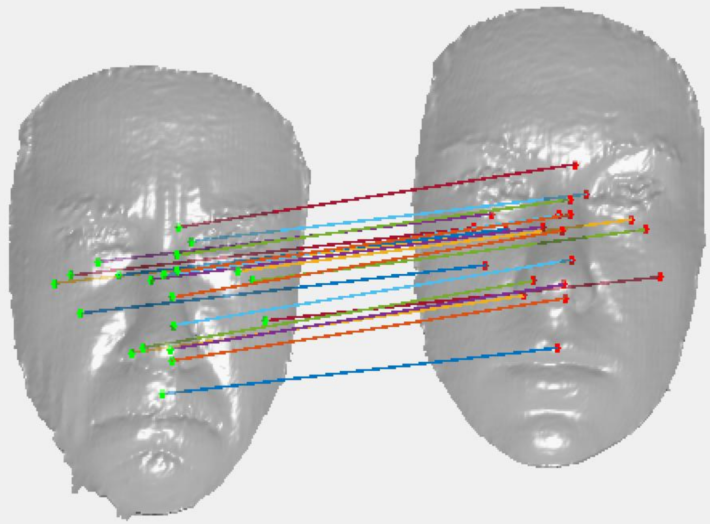

AMCAP - F:\
File Devices Options Capture Photo Help


Visible
Near-Infrared
Thermal
FLIR
(a)
(b)
(c)
(d)
(e)

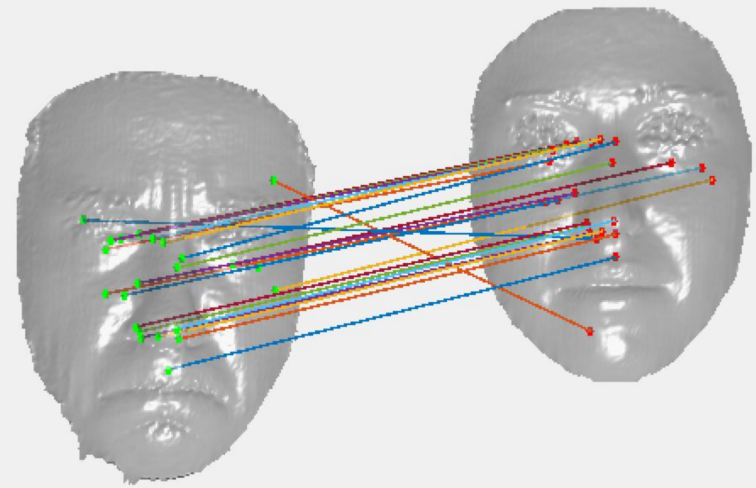

frame0
Real

三维人脸识别

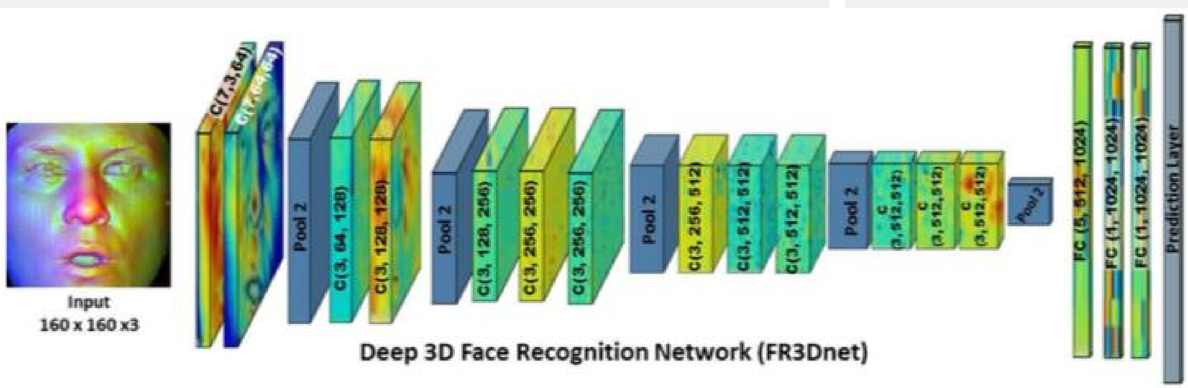




Same person



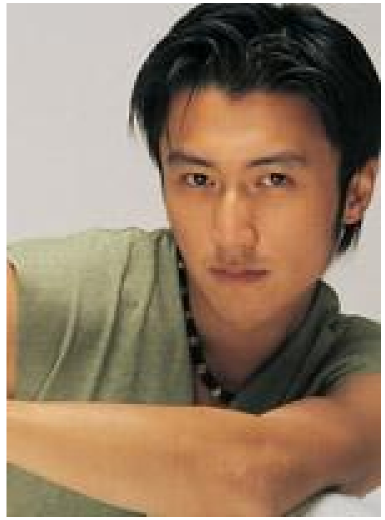
Different persons



Deep 3D Face Recognition Network (FR3Dnet)

Shenzhen University

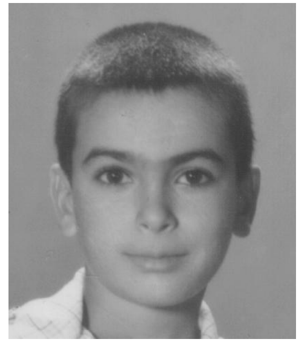
年龄变化

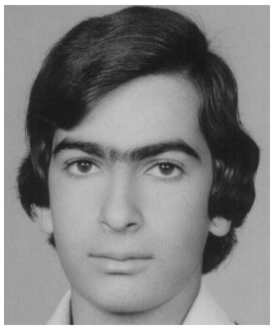
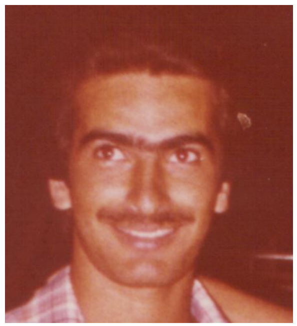
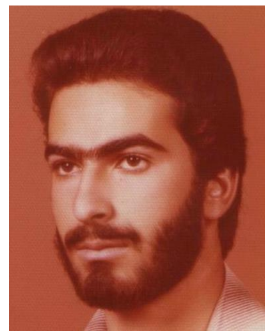







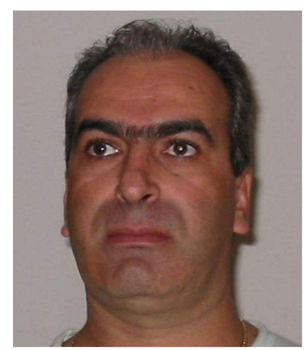




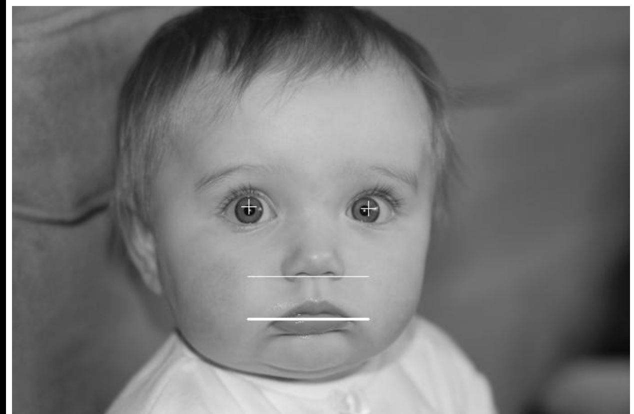





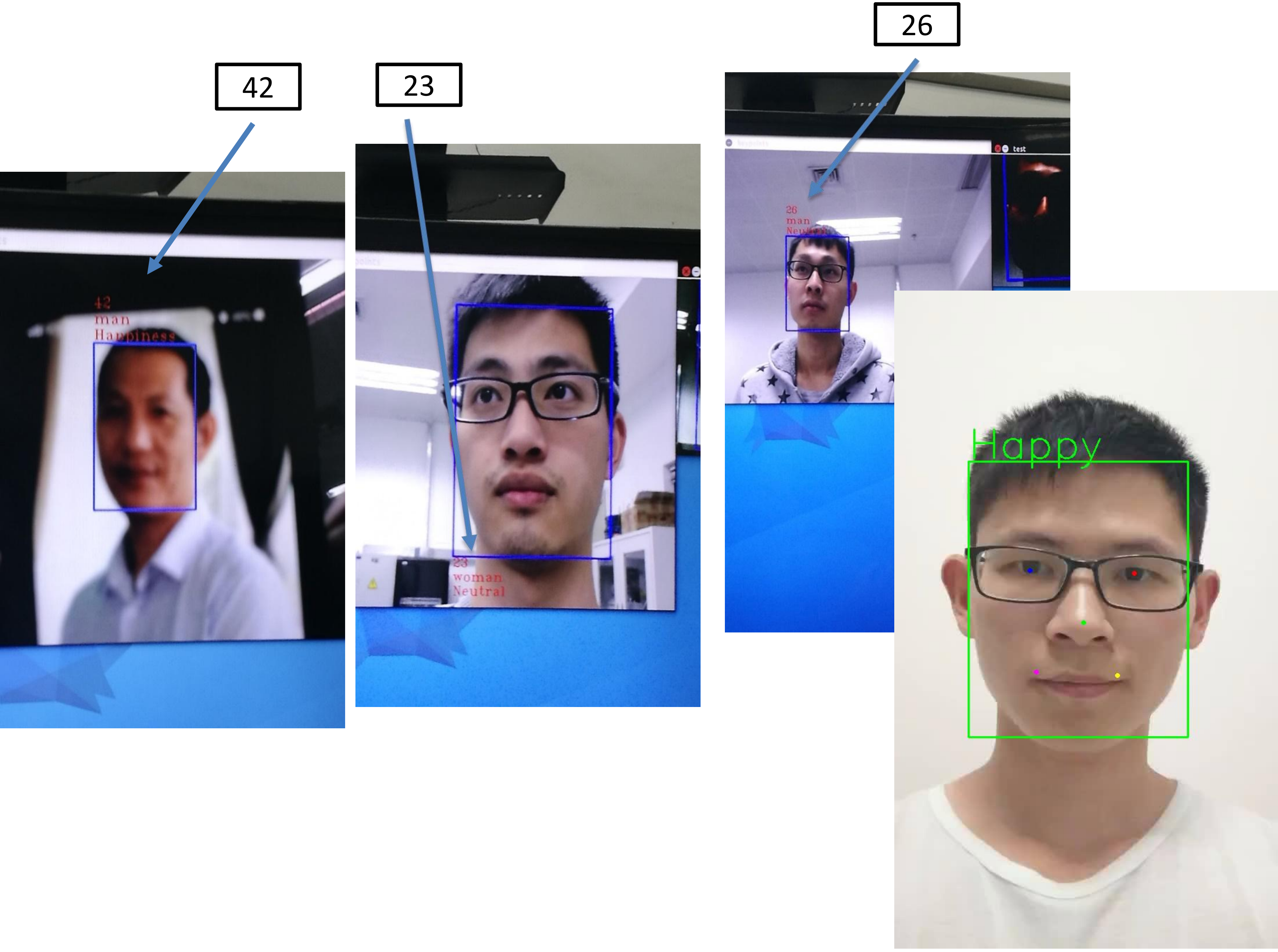




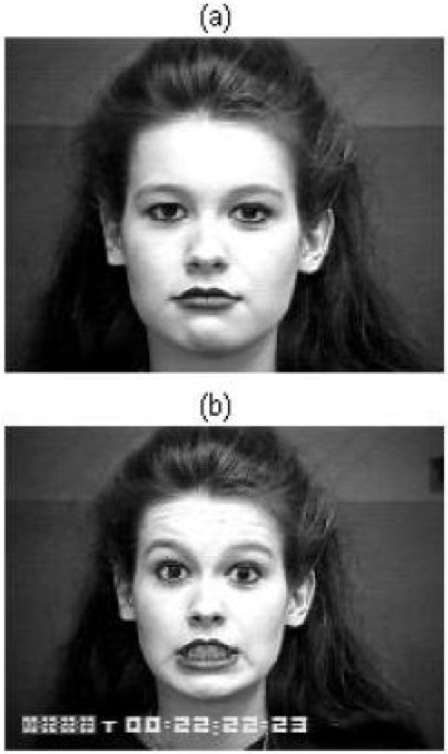


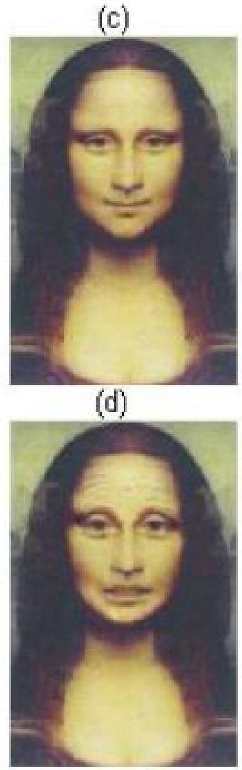
年龄判别

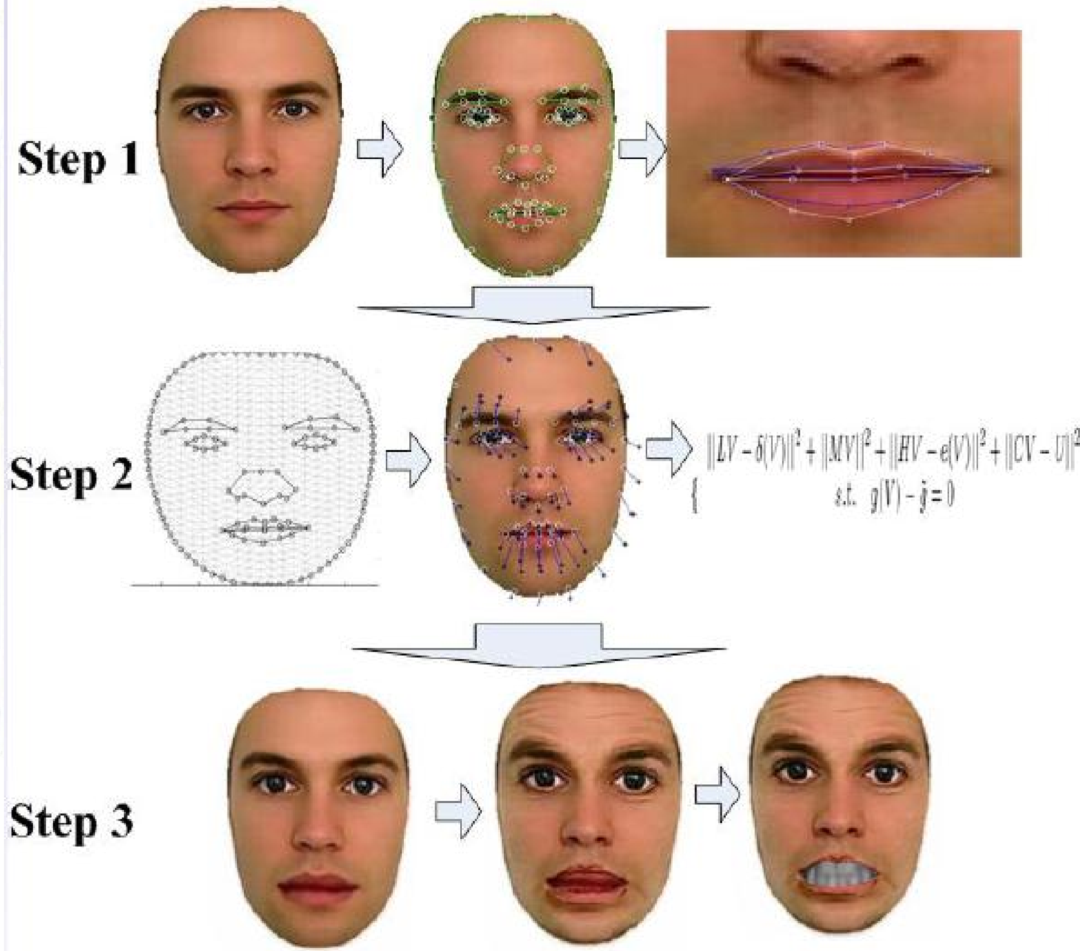


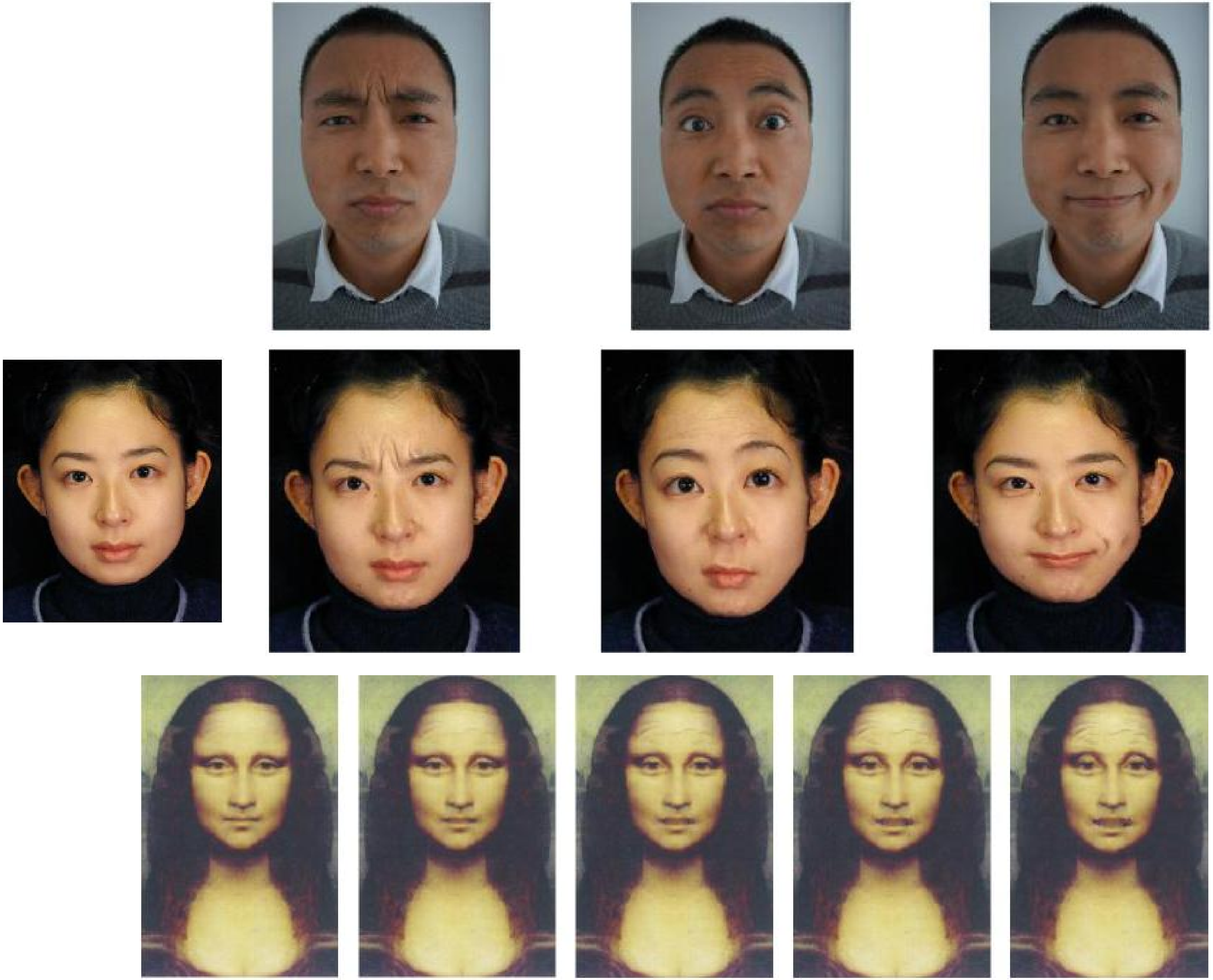

42
23
26
42
man
Happiness
23
woman
Neutral
26
man
Happy

表情克隆

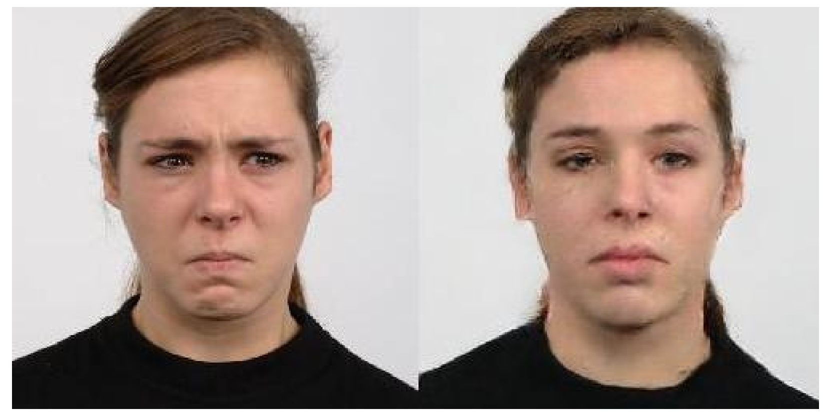






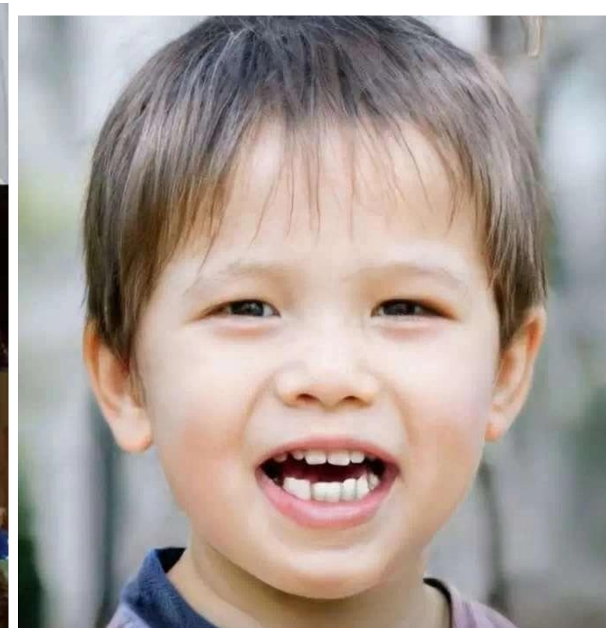


去表情

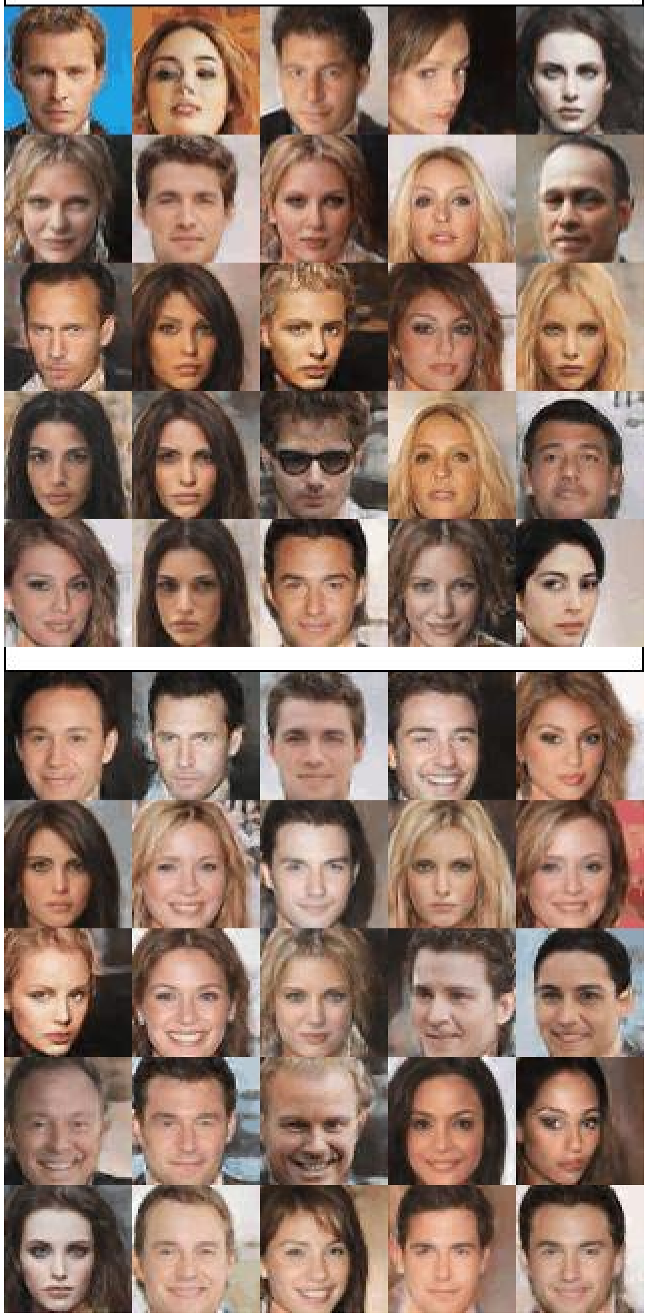


Big GAN

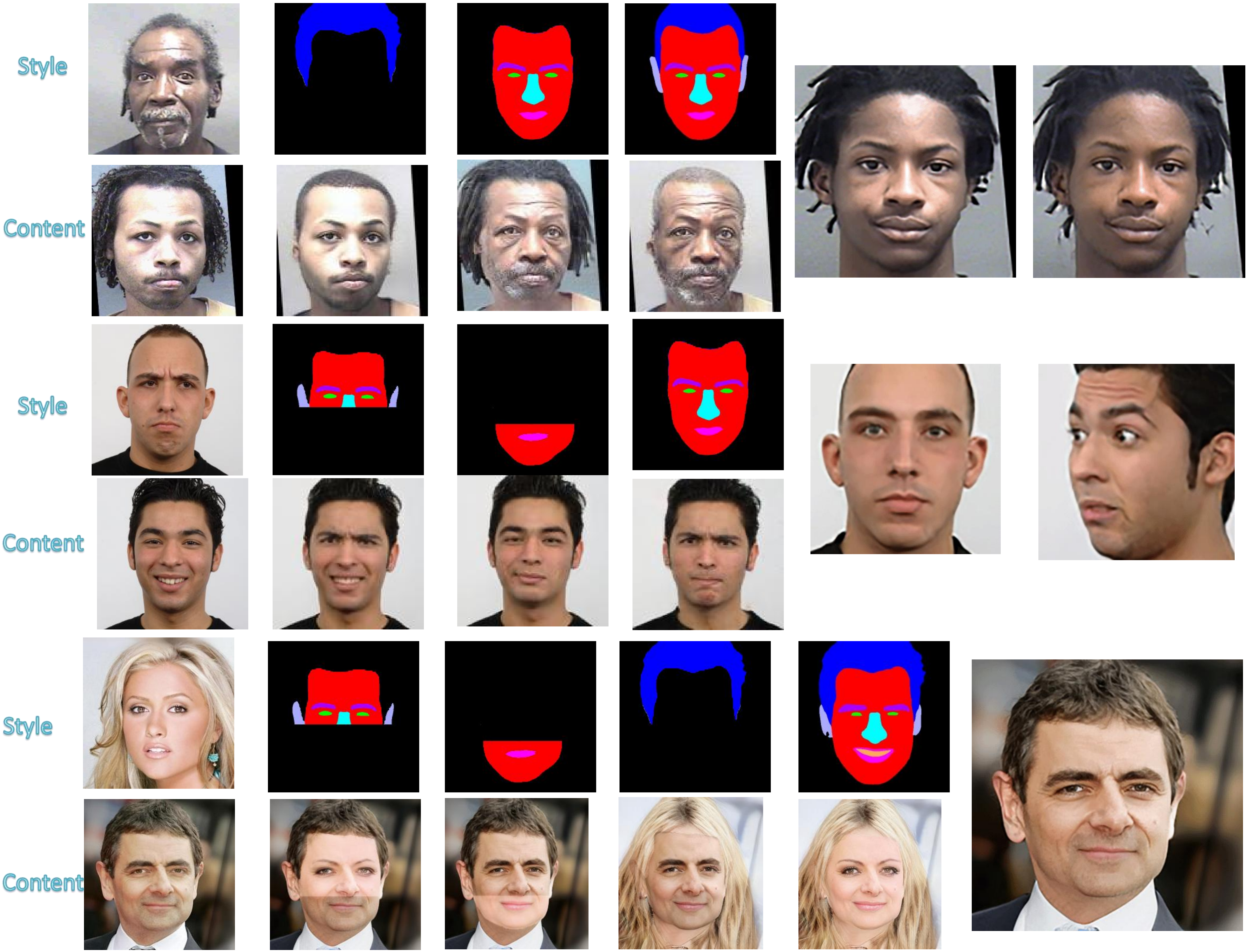
2018.10



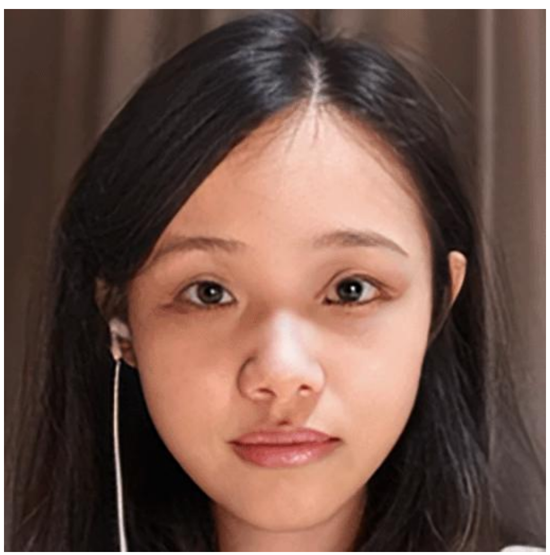
Make face smile



Make mustache on face


Style
Content
Style
Content
Style
Content



卡通漫画


Reference